Mitigating Temporal Domain Disparities for Cross-Temporal Remote Sensing Semantic Change Detection Using Vector Parcel Data and Contrastive Learning

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***Abstract*—Most existing change detection (CD) methods assume that multitemporal remote sensing images are acquired under similar seasonal conditions. When this assumption is violated, significant phenological variations introduce large appearance discrepancies that severely degrade detection accuracy. Moreover, most deep-learning-based CD approaches rely solely on raster imagery and neglect historical vector parcel data, resulting in limited training samples and poor generalization in real-world applications. To address these challenges, this article proposes a vector-guided cross-temporal semantic CD (SCD) framework that mitigates temporal appearance discrepancies through contrastive feature alignment. A boundary-preserved superpixel segmentation and vector parcel-constrained adaptive cropping strategy is employed to automatically generate vector-aligned training samples, followed by a category-constrained clustering scheme to suppress label noise. A hybrid cross-temporal learning framework is then designed to reduce feature-level discrepancies between temporally mismatched images using Fourier-enhanced representations and contrastive supervision. Semantic changes are finally identified at the vector parcel level through rule-based reasoning. Experiments on two real-world regional datasets demonstrate that the proposed method achieves robust parcel-level SCD under significant seasonal variations, outperforming representative state-of-the-art approaches and exhibiting strong generalization capability.**

***Index Terms*—Contrastive learning (CL), cross-temporal remote sensing, feature alignment, semantic change detection (CD), vector parcel data.**

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I. Introduction

ACCURATELY delineating surface changes with regular boundaries and identifying their semantic categories from satellite imagery remains a major technical challenge, particularly when substantial spectral and seasonal (phenological) differences exist between images acquired at different times [1]. Variations in illumination, atmospheric conditions, vegetation states, and sensor geometry often introduce significant radiometric inconsistencies and geometric misalignments in high-resolution multitemporal imagery, as illustrated in Fig. 1, which severely degrade change detection (CD) performance.

Reliable extraction of land-cover change information is fundamental to global change studies, land resource management, and ecosystem monitoring. However, discrepancies in acquisition time, satellite attitude, and viewing geometry frequently result in false alarms when existing CD methods are applied to high-resolution imagery [2]. Mitigating the interference caused by seasonal spectral variations and other external factors has therefore become a prerequisite for robust CD [3]. Meanwhile, high-quality training samples are costly to obtain, and labels from different temporal phases are rarely transferable [4], motivating increasing interest in automated sample generation and feature-level domain alignment strategies, which aim to reduce distributional discrepancies without requiring explicit target-domain retraining [5], [6].

Large appearance variations ("style shifts") and complex background patterns across bi-temporal or cross-regional images hinder the generalization ability of CD models. As a result, achieving rapid and accurate extraction of cross-temporal change information has become an urgent priority. Existing approaches for extracting change information can be broadly categorized into two groups. The first involves crowdsourced data, such as comparing OpenStreetMap (OSM)

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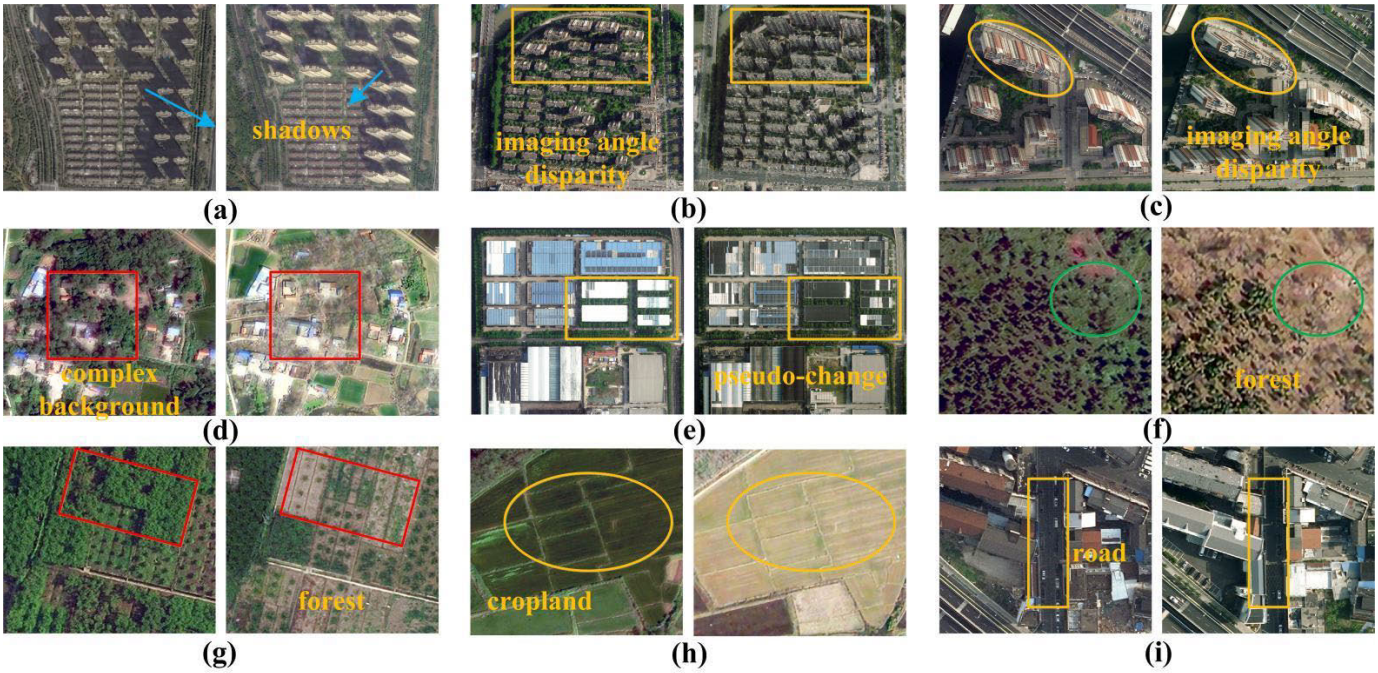


Fig. 1. Challenges in cross-temporal image CD. The illustrated challenges include (a) shadow interference caused by illumination variations; (b) and (c) intraclass variability of buildings arising from differences in imaging conditions; (d) complex background clutter; and (e) tonal differences at identical locations, resulting in pseudo-changes. In addition, (f)–(h) illustrate substantial spectral and textural variations induced by vegetation growth and seasonal transitions, while (i) shows subtle misregistration and shape discrepancies along road boundaries between the image pair.

with remote sensing imagery to detect changed objects. The second comprises domain alignment–based methods, which aim to learn domain-invariant representations by explicitly reducing feature distribution discrepancies across temporal or spatial domains, often through auxiliary constraints or proxy learning objectives. A special challenge arises from time-related "false changes" caused by illumination and seasonal variations: the same object may exhibit stark visual differences between cloudy and sunny conditions, a phenomenon referred to in this article as cross-temporal shift. Importantly, this shift pattern often appears consistently across entire datasets [7].

Recent studies have made significant progress in enhancing CD robustness. To alleviate false changes induced by acquisition-time inconsistencies, Lu et al. [8] proposed an object-oriented fusion model integrating Landsat and MODIS data. Toure et al. [9] combined object-based approaches with retrospective analysis for robust multiresolution CD, while Jing et al. [10] incorporated image segmentation techniques to exploit the spatial, spectral, and multitemporal properties of objects. These works demonstrate the advantages of object-oriented strategies for interpreting fine-scale, complex structures in high-resolution imagery.

Feature extraction plays a central role in object-oriented CD. Deep learning models, particularly convolutional neural networks (CNNs), have been widely adopted due to their strong feature-learning and representation capabilities. Several improved CNN-based methods have been proposed, including the full-scale difference feature fusion network (FDFF-Net) by Gu et al. [11] and the cross-attention-guided group aggregation network (CAGNet) for cultivated land CD. Further advancements, such as large-kernel hybrid CNNs [12] and adaptive Fourier convolution networks (AFCNets) [13], have continued to refine this research line. Despite these achievements, CNN-based methods still struggle to capture long-range dependencies and suffer from over-smoothing effects.

With the emergence of transformers, hybrid CNN–transformer architectures have become a major research focus, achieving substantial improvements in CD. Representative examples include the BIT_CD model [14], which leverages transformer modules to enhance spatio-temporal modeling, and ChangeFormer [15], which employs a hierarchical transformer encoder and MLP decoder for multiscale feature extraction. Beyond CNNs and transformers, state space models (SSMs), such as Mamba [16], have recently demonstrated exceptional performance in various domains due to their efficient selective state modeling. However, their potential for remote sensing image change characterization (RSICC) remains largely underexplored.

Although deep-learning-based CD methods offer strong representational capabilities, they generally require large annotated datasets, and obtaining high-quality labels is time-consuming and labor-intensive. To mitigate this limitation, various automatic sample-generation approaches have been explored. Liu et al. [17] utilized spectral indices and spectral-angle matching to generate training samples for crop extraction. Luppino et al. [18] computed unchanged regions using affinity matrices as pseudo-labels for CD. Lv et al. [19] introduced an iterative sample-enhancement strategy for deep CD models, while Liu et al. [20] proposed a noise-driven deep learning method using low-resolution prior data to address cross-resolution semantic errors. Nevertheless, these approaches often assume similar data distributions between

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training and testing domains, an assumption that rarely holds in practice. When multitemporal images are acquired at inconsistent times, domain shifts severely degrade model generalization. Some studies reduce this discrepancy via image regression to a common domain [21], but such methods are mainly suitable for low-resolution imagery. Other works have explored tone transfer and knowledge mixing for building mapping [6], or self-supervised pretraining frameworks (e.g., RECM and SITS-MoCo) to reduce annotation requirements [22], [23].

In parallel, DA, style transfer, and transfer learning have been investigated to alleviate annotation scarcity. Dong et al. [24] explored single-domain generalization for hyperspectral imagery, while Su et al. [25] introduced a lightweight adapter for few-shot cross-domain segmentation. Toldo et al. [26] proposed a style-learning framework for knowledge transfer across tasks and domains. Other works transformed bi-temporal imagery into a shared domain to suppress differences for landslide detection [27]. Multitask pretraining (MTP) has also gained attention in semantic segmentation, classification, and CD [28]. However, despite these advances, most existing research has focused on spatial-domain shifts across geographic regions, with comparatively little attention given to temporal domain shifts. In contrast to spatial-domain shifts, temporal domain shifts induced by seasonal and phenological variations exhibit dataset-level consistency but object-level appearance discrepancies, posing unique challenges that are not fully addressed by existing DA frameworks.

Most existing CD studies rely primarily on image data, whereas vector parcel data, which contain rich boundary and semantic information, remain underutilized despite their strong potential to support segmentation, annotation, and semantic change analysis. Several studies have explored this direction. For example, a contrastive learning (CL) framework was developed in [29] to extract building footprints from historical maps and modern imagery. Zhang et al. [30] introduced vector parcel data and the isolation forest algorithm for CD, and Wei et al. [31] proposed a texture-based outlier index for detecting changes from single-temporal imagery and historical vector parcel data. However, the reliance on texture limits general applicability. More recent work incorporated POI data and OSM annotations for global land-use mapping through a POI-prompted deep learning framework [32]. Despite these efforts, the effective integration of vector parcel data with multitemporal imagery remains largely underexplored in practical applications.

In parallel, large-scale models have significantly advanced remote sensing image interpretation. Representative approaches, such as SkySense [33], leverage large-scale architectures and pretraining strategies to enhance feature representation and generalization across diverse scenarios. Meanwhile, recent semantic CD (SCD) methods, including TSCCD [34], have investigated temporal modeling and cross-domain adaptation (DA) to mitigate appearance discrepancies. Nevertheless, these approaches seldom exploit structured geographic information, such as vector parcel data, which is often readily available in operational settings. Consequently, although existing CD methods benefit from either vector-guided learning or large-model architectures, they continue to face challenges related to limited annotations, label noise, and complex cross-temporal variations.

To address these issues, this article proposes a novel DA framework for SCD in cross-temporal remote sensing imagery, leveraging vector parcel data and CL. The framework tackles two core challenges: 1) automatic acquisition of high-quality training samples and 2) knowledge transfer when pre-event and postevent images are acquired at inconsistent times.

The main contributions of this study are summarized as follows.

1) *Cross-Temporal Generalization-Oriented SCD Framework:* We propose a vector-guided SCD framework that explicitly addresses temporal appearance discrepancies rather than conventional spatial-domain shifts. By integrating historical vector parcel data with CL, the framework enables robust parcel-level CD under heterogeneous acquisition times, thereby enhancing applicability in real-world, long-term monitoring scenarios.
2) *Automated Vector Parcel-Constrained Sample Construction and Noise Suppression for Practical SCD:* To address the lack of reliable training samples in operational CD, we develop a boundary-preserved superpixel segmentation and vector parcel-constrained adaptive cropping strategy for automated, vector-aligned sample generation. Furthermore, a category-constrained clustering-based denoising scheme is introduced to suppress label noise, alleviating a critical bottleneck that limits the generalization performance of existing CD methods in practice.
3) *Hybrid Supervised Cross-Temporal Learning Mechanism for Reducing Temporal Feature Discrepancies:* We design a hybrid supervised cross-temporal learning framework (HCLF) that jointly leverages supervised, self-supervised, and CL, complemented by Fourier-enhanced representations, to mitigate feature-level discrepancies between temporally mismatched images. A rule-based change decider further incorporates domain knowledge to support reliable and interpretable parcel-level change inference.

Compared with existing deep-learning-based CD methods as follows.

1) *Model Robustness:* Most existing CD methods rely on direct image comparison and are therefore sensitive to noise and resolution differences. In this study, CD is reformulated as a vector-guided semantic inference problem, in which semantic categories are first inferred and then compared using vector parcel attributes. On this basis, the HCLF framework incorporates Fourier-domain transformations to mitigate cross-temporal discrepancies and enhance model robustness.
2) *Sample Generation:* Unlike self-supervised [6] or model-transfer approaches, the proposed method incorporates vector parcel data as prior knowledge and introduces an automated sample generation framework.

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This design substantially improves training-sample quality and better supports practical CD workflows.

II. Related Work

A. Deep-Learning-Based CD

With the rapid development of deep learning, traditional supervised CD methods have evolved toward increasingly deep and complex architectures. Deep-learning-based CD approaches can be broadly categorized into classification-based and metric-based methods, depending on how the change map is produced. In classification-based methods, bi-temporal features are fused at specific stages, and the final change map is generated by a classifier at the network head. For example, Ding et al. [35] proposed the bi-temporal semantic reasoning network (Bi-SRNet), which integrates a cross-temporal semantic reasoning attention module to enhance semantic consistency between predictions from different temporal inputs. Chen et al. [36] introduced the feature constraint CD network (FCCDN), which robustly fuses bi-temporal features to achieve accurate semantic segmentation. In addition to CNN-based architectures, transformer-based models have also demonstrated strong performance in CD tasks [14], [37].

Metric-based CD methods transform paired input images into a shared feature space and subsequently derive the change map through a distance metric. For instance, Shi et al. [38] introduced the deep siamese attention metric network (DSAMNet), which employs multiple ResNet modules to extract multiscale features. Zhao et al. [39] proposed a near-symmetric dual-branch neural network whose coupled layers ensure comparable feature transformations for the two temporal images. Bandara and Patel [40] designed an unsupervised CD method based on deep metric learning, which iteratively refines detection results by minimizing a customized similarity-dissimilarity loss.

Beyond this dichotomy, CD methods may also be classified by granularity into pixel-based, object-based, and scene-based approaches. Among these, object-based CD has demonstrated notable progress due to its ability to exploit spatially coherent superpixels, which are relatively homogeneous groups of pixels representing meaningful entities. Object-based CD is less sensitive to observation conditions and effectively mitigates random noise and seasonal variation, making it suitable for applications such as building detection [10]. However, its performance depends heavily on selecting appropriate superpixel segmentation thresholds, and segmentation quality directly affects detection accuracy.

In object-based CD, the choice of object-generation strategy strongly influences performance, as consistent object boundaries across temporal phases are essential. Clustering-based segmentation approaches conceptualize superpixel generation as a clustering problem, providing advantages in efficiency, compactness, and controllability. Among them, simple linear iterative clustering (SLIC) is one of the most widely adopted methods due to its simplicity and computational efficiency. For example, Shi et al. [41] used SLIC-based covariance feature extraction to obtain robust structural features, reducing sensitivity to speckle noise and avoiding redundant superpixel representations. Zhan et al. [42] combined a bilinear CNN with SLIC to reduce annotation effort while improving accuracy. Zhang et al. [43] developed the end-to-end superpixel-enhanced CD network (ESCNet), which accurately localizes changed regions. Since superpixels cannot be directly processed by deep networks, they are typically vectorized and cropped or padded into image patches for subsequent learning. Collectively, these studies highlight significant advances in object-level CD facilitated by superpixel segmentation.

Scene-based CD focuses on changes at the entire-scene level rather than object-level alterations. For example, transitions such as building completion, newly constructed roads, or increased green space may not alter the fundamental scene type, even if individual objects change. Scene-based CD can distinguish object categories and selectively extract relevant changes, improving interpretability and semantic understanding. However, this approach requires building specialized models or rule sets for various scenarios, creating challenges in complex environments.

Despite notable advancements, accurately extracting discriminative features across diverse change scenarios, suppressing false alarms, and improving robustness in high-resolution imagery remain open challenges.

B. Sample Collection and Evolution of Deep Learning Architectures

The limited availability of high-quality annotated samples remains a major bottleneck for the widespread adoption of deep learning in remote sensing. As a result, automatic sample collection and label-efficient learning strategies have attracted increasing attention. Representative studies have explored pseudo-label generation and weakly supervised learning to alleviate annotation costs. For example, Li et al. [44] proposed a pseudo-label-based strategy to enhance zero-shot scene classification, while Shao et al. [45] introduced a few-shot pretraining framework for open-pit mine extraction under limited supervision. In addition, Li et al. [46] combined CNN and transformer architectures to update high-resolution land-cover maps using historical low-resolution labels, effectively reducing manual annotation effort.

Beyond label-efficient learning, several studies have investigated automated sample generation guided by spectral or temporal characteristics. Phenology-driven approaches have been employed for large-scale crop mapping, while multi-source data integration strategies combining LiDAR or SAR with optical imagery have shown effectiveness in agricultural and land-cover applications [47]. Cross-scale sample transfer methods have also been explored to generate reliable high-resolution samples from coarse labels.

Meanwhile, advances in deep learning architectures, including CNN variants and transformer-based models, have significantly improved performance on remote sensing benchmarks. However, challenges such as scale variability, background complexity, and limited generalization remain. To address these issues, researchers have focused on enhancing feature representation through architectural innovations, such as capsule-based networks [48], bilinear pooling

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mechanisms [49], and multiscale attention frameworks [37]. More recently, frequency-domain modeling has been introduced to jointly capture global context and fine-grained details beyond conventional self-attention mechanisms [13].

Another closely related research direction concerns robust learning under noisy labels. Typical approaches include progressive sample selection and iterative label refinement [50], as well as uncertainty-aware and metric-learning-based strategies for identifying unreliable samples [51]. Although these methods demonstrate promising performance, they often rely on complex noise modeling or aggressive sample filtering, which may reduce sample diversity or limit scalability in real-world scenarios.

Overall, existing studies highlight the importance of automated sample construction and robust feature learning. However, effectively integrating high-quality sample generation with generalizable cross-temporal representation learning remains an open challenge in practical CD applications.

C. DA and Feature Alignment in Remote Sensing

In remote sensing applications, deep learning models often suffer from performance degradation due to distributional discrepancies between training and testing data. While DA traditionally refers to transferring a trained model from a label-rich source domain to a label-scarce or unlabeled target domain, recent studies increasingly emphasize feature-level domain alignment, where representations from different domains are encouraged to follow similar distributions during joint training. This paradigm is particularly suitable for cross-temporal scenarios, in which temporal appearance variations are systematic, while explicit target-DA data are typically unavailable.

Many DA approaches adapt the entire model to the target domain, which may lead to overfitting when target data are limited or highly variable. To address this issue, several studies have explored alternative strategies that avoid full model adaptation. Yuan et al. [52] proposed a diffusion-based framework (RSFSG) to generate controllable synthetic samples, while Su et al. [25] introduced compact adapter modules to correct target-domain features without modifying the entire network. Other works incorporated environmental constraints or multitask learning [27] to improve cross-domain robustness.

More recently, feature-alignment-oriented methods have shown strong performance in heterogeneous and cross-temporal CD. C3D [53] combines dual image translation with CL, while MDCTNet employs multidomain constrained translation with high-frequency preservation. To further model cross-domain dependencies, MBATrans [54] establishes semantic relationships across domains, and [55] exploits Fourier-based frequency consistency for multimodal CD. These studies demonstrate that feature distribution alignment is an effective means of mitigating temporal domain shifts when the semantic label space remains consistent.

III. Methodology

This section introduces the proposed DA-based feature alignment framework for cross-temporal remote sensing SCD. As illustrated in Fig. 2, the workflow comprises five sequential stages: 1) vector-boundary-constrained superpixel segmentation of bi-temporal imagery; 2) superpixel-guided adaptive sample construction; 3) training-sample denoising via category-constrained clustering; 4) cross-temporal feature alignment and classification using the proposed HCLF; and 5) parcel-level postclassification SCD. Specifically, the historical vector map at time T1 is employed as a geometric prior to guide boundary-preserved superpixel segmentation for both T1 and T2 images, thereby ensuring spatial consistency across temporal phases. Based on these superpixels, vector-aligned image patches are adaptively cropped to construct initial training and testing datasets. To suppress label noise induced by temporal discrepancies, a pretrained vision transformer (ViT) combined with category-constrained density peak clustering (C^2-DPC) is applied to refine the training samples. The resulting clean samples are then input into HCLF to learn temporally invariant feature representations through joint supervised learning, CL, and feature-level distribution alignment. Finally, SCD is performed by overlaying the classification results with the original vector parcel units and applying rule-based change criteria.

A. Image Superpixel Segmentation and Automated Sample Set Construction

1) Vector-Boundary-Constrained Superpixel Segmentation: Among existing superpixel segmentation algorithms, SLIC is widely adopted due to its computational efficiency and its ability to generate compact superpixels by jointly considering spectral similarity and spatial proximity. However, SLIC does not explicitly enforce object-level boundary constraints and may produce superpixels that cross land-cover boundaries, particularly in complex scenes. To address this limitation, we adopt the BPSLIC algorithm, in which superpixel generation is strictly constrained by vector parcel boundaries (i.e., polygon-based geographic units).

In this study, the vector map at time T1 is employed as a geometric boundary prior to guide superpixel segmentation for both T1 and T2 images. Each vector parcel is treated as an independent segmentation region, within which superpixels are generated separately. This design is motivated by the observation that, although spectral and textural characteristics may vary significantly between T1 and T2 due to phenological changes, the topological structure and spatial extent of standardized geographical entities remain relatively stable over the considered temporal interval. Therefore, using the T1 vector map provides consistent spatial partitioning while avoiding semantic interference from temporal appearance variations.

The detailed procedure is as follows.

Step 1 (Initialization of Cluster Centers): Given a vector parcel region and a desired superpixel size, seed points are iteratively placed within the polygon such that the distance from each seed point to the region boundary and to other seed points is maximized, ensuring a uniform spatial distribution. The seed point P is determined as

$$P = \arg\max_{x} (\|x - y\|_2) \tag{1}$$

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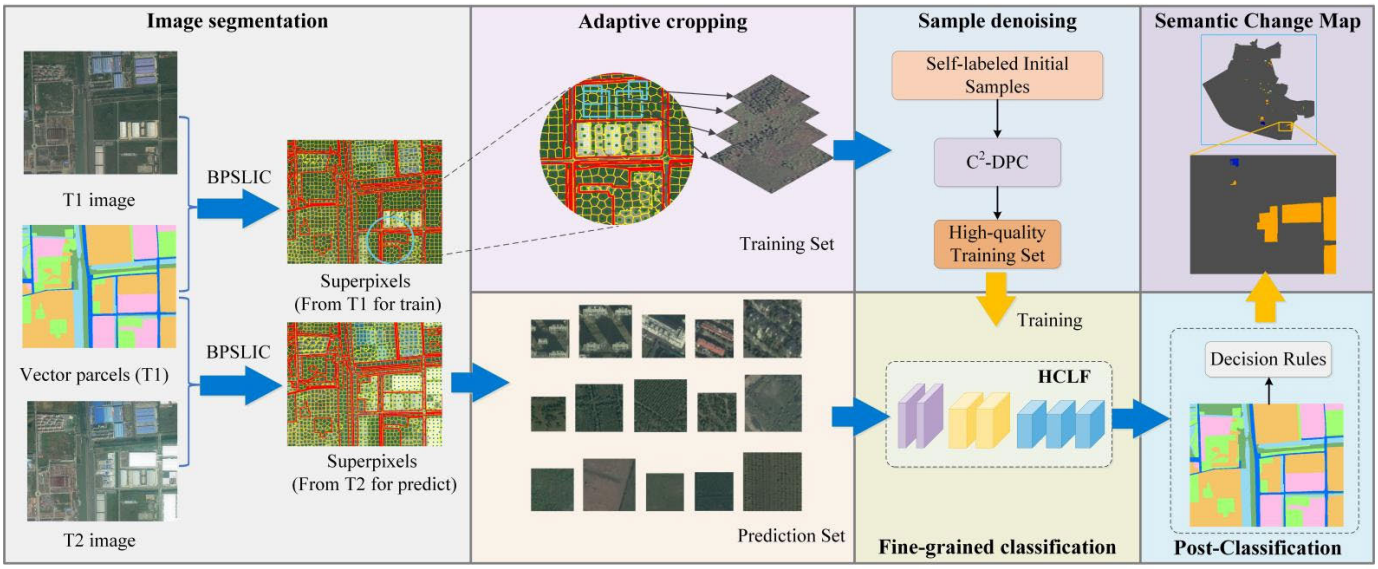


Fig. 2. Overview of the proposed DA-based feature alignment framework for cross-temporal SCD.

where y denotes points on the polygon boundary and the coordinates of already placed seed points, and x represents candidate coordinates of unplaced seed points. Since vector parcels vary in shape and size, the number of superpixels within each polygon is adaptively determined. Let N denote the total number of pixels within a polygon region and C_s represent a predefined segmentation size standard. Assuming each superpixel is approximately square with side length $S = m \times C_s$, where m is a scaling constant, the number of superpixels K in the polygon is calculated as

$$K = \frac{N}{(m \times C_s)^2}. \tag{2}$$

This adaptive strategy ensures that generated superpixels remain compact and approximately uniform across polygons of different scales.

Step 2 (Distance Computation Within Polygon Constraints): For each pixel $p_i^{\boldsymbol{I}}$ in the image $\boldsymbol{I}$, distances to cluster centers within a $2S \times 2S$ neighborhood are computed by jointly considering spatial and spectral information

$$d_{spatial}\left(p_i^{\boldsymbol{I}}, p_j^{\boldsymbol{I}}\right) = \sqrt{\left(r_i^{\boldsymbol{I}} - r_j^{\boldsymbol{I}}\right)^2 + \left(c_i^{\boldsymbol{I}} - c_j^{\boldsymbol{I}}\right)^2} \tag{3}$$

$$d_{spectral}\left(p_i^{\boldsymbol{I}}, p_j^{\boldsymbol{I}}\right) = \sqrt{\left(x_i^{\boldsymbol{I}}\right)^2 + \left(x_j^{\boldsymbol{I}}\right)^2} \tag{4}$$

$$d_{sum}\left(p_i^{\boldsymbol{I}}, p_j^{\boldsymbol{I}}\right) = \omega \times d_{spatial}\left(p_i^{\boldsymbol{I}}, p_j^{\boldsymbol{I}}\right) + (1-\omega) \times d_{spectral}\left(p_i^{\boldsymbol{I}}, p_j^{\boldsymbol{I}}\right) \tag{5}$$

where $(r_i^{\boldsymbol{I}}, c_i^{\boldsymbol{I}})$ and $(r_j^{\boldsymbol{I}}, c_j^{\boldsymbol{I}})$ denote the spatial coordinates of pixels $p_i^{\boldsymbol{I}}$ and $p_j^{\boldsymbol{I}}$, respectively, and $x_i^{\boldsymbol{I}}$ and $x_j^{\boldsymbol{I}}$ represent their corresponding spectral feature vectors. The weighting parameter ω controls the relative contributions of spatial and spectral distances. In practice, ω is adaptively adjusted according to the terrain characteristics of the target region. Specifically, it is typically set to 0.5 in urban areas, increased to 0.7 in transitional regions such as forest–farmland boundaries to emphasize spatial consistency, and reduced to 0.3 in homogeneous farmland areas to prioritize spectral similarity.

Step 3 (Boundary-Constrained Clustering and Update): Each pixel is assigned to the nearest cluster center within the same vector parcel, ensuring that no superpixel crosses polygon boundaries. Cluster centers are updated by computing the mean feature vector of assigned pixels, and the pixel closest to the mean is selected as the new center. The process iterates until convergence, yielding the final superpixel set $SP = \{sp_1, sp_2, \ldots, sp_K\}$.

2) Automated Sample Set Construction Supported by Adaptive Cropping: Based on the superpixel segmentation results shown in Fig. 2, vector parcel-constrained adaptive cropping is applied to generate scene-level samples. High-quality training samples are essential for accurate semantic interpretation, yet their manual collection is costly and labor-intensive. Moreover, class labels cannot be directly transferred across temporal phases, further complicating sample acquisition. In contrast, natural resource management agencies typically maintain extensive survey datasets in the form of vector parcels, which contain rich boundary and attribute information.

To exploit these advantages, we adopt a superpixel-guided adaptive cropping strategy [4] to construct an initial sample set. Unlike fixed-size cropping windows, which often introduce boundary contamination, this approach dynamically adjusts the crop size based on the distance to the vector parcel boundary, thereby improving sample purity near object edges. Specifically, given an initial cropping size C_0, the effective cropping size C is defined as

$$C = \begin{cases} C_0, & d_b > C_0 \\ d_b, & d_b \le C_0 \end{cases} \tag{6}$$

where d_b denotes the shortest distance from the cropping center to the vector parcel boundary.

Because vector survey data are collected overextended production cycles, registration errors and annotation

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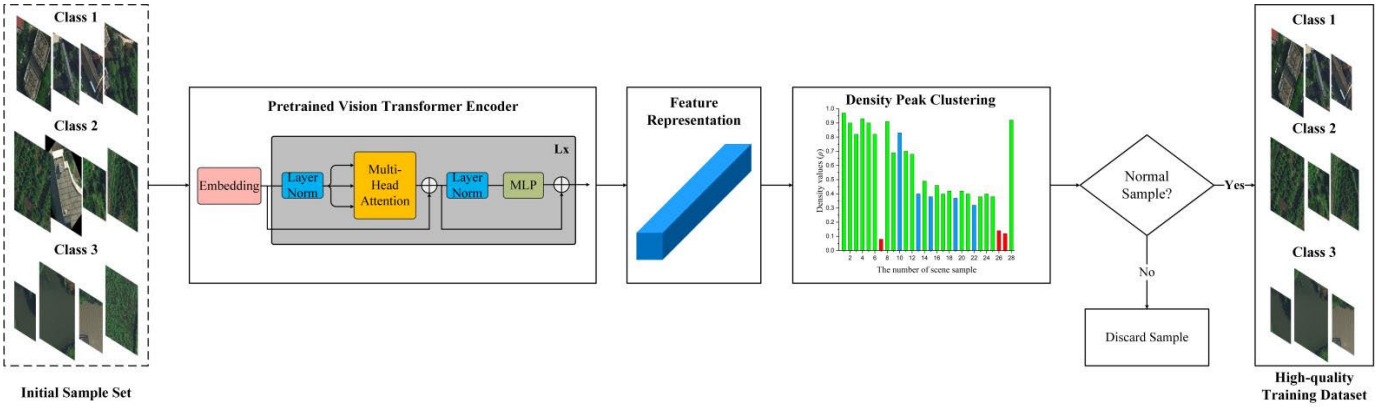


Fig. 3. Proposed C^2-DPC denoising framework. The initial sample labels are directly inherited from the vector parcel data.

inconsistencies may lead to discrepancies between vector attributes and actual land-cover conditions in high-resolution imagery. Consequently, automatically generated parcel-aligned image samples inevitably contain label noise, which can degrade model training. To address this issue, we introduce a C^2-DPC denoising framework, illustrated in Fig. 3. Since scene-level image patches must first be represented numerically, a pretrained ViT is employed to extract feature vectors for each sample. These features are then input into the C^2-DPC algorithm to identify and remove noisy samples, yielding a refined and reliable training set.

B. Hybrid Supervised Cross-Temporal Learning Framework

Significant intraclass variations frequently arise in cross-temporal remote sensing imagery due to seasonal changes, illumination differences, and phenological effects. These variations often lead to degraded performance for conventional supervised learning methods, particularly when training samples are limited or temporally inconsistent. In contrast, CL alleviates the reliance on precise annotations by learning discriminative representations through similarity relationships, making it well-suited to scenarios with scarce labels or substantial label noise.

Depending on the construction of sample pairs, CL can be broadly categorized into instance-based (label-free) and label-based (label-constrained) paradigms. Instance-based CL forms positive pairs from different augmented views of the same image, while label-based CL treats samples sharing the same semantic label as positives. Although CL has demonstrated strong representation-learning capability, conventional CL pipelines typically follow a two-stage strategy consisting of self-supervised pretraining and task-specific fine-tuning. When training and testing data originate from different temporal domains, such strategies often suffer from distribution mismatch, leading to limited generalization to target-domain imagery.

To address this issue, we propose an HCLF that performs temporal DA through explicit feature alignment. It should be noted that the proposed framework does not perform explicit source-to-target model adaptation; instead, it achieves temporal DA by aligning feature distributions across time through CL and MMD-based regularization during joint training. By jointly exploiting labeled samples from the source temporal domain and unlabeled samples from the target temporal domain, HCLF explicitly models cross-temporal distribution shifts during representation learning. Unlike conventional semi-supervised or CL approaches that focus solely on the source-domain distribution, HCLF incorporates both the training (source) and prediction (target) domains into a unified learning pipeline. As illustrated in Fig. 4, self-supervised CL and supervised classification are integrated into a single-stage framework. Labeled and unlabeled samples are fed into the network simultaneously, allowing supervised signals to guide category discrimination while unlabeled target-domain samples regularize feature representations toward temporal invariance. After training, only the weights of the supervised branch are retained for inference.

For cross-temporal scene classification, we further introduce a spatio-temporal category-constrained contrastive strategy. Instead of constructing positive pairs through random augmentation, two images acquired at different times but corresponding to the same geographical location are treated as positive pairs, while samples from different locations within the same batch serve as negative pairs. This design encourages the model to suppress temporal appearance variations and focus on stable structural characteristics, such as spatial layout and texture, thereby directly addressing the core challenge of cross-temporal domain shift.

Within the CL component, we adopt a network architecture inspired by SimCLR [56]. For any unlabeled pre-event and postevent image pair from the same location, two augmented views $(\tilde{t}_i, \tilde{t}_j)$ are generated to form a positive pair. These views are transformed into the feature space through a backbone encoder and a projection head, formulated as

$$h_i = f\left(\tilde{t}_i\right) \tag{7}$$

$$z_i = MLP\left(h_i\right) \tag{8}$$

where $f(\cdot)$ denotes the backbone feature extractor and $MLP(\cdot)$ is the projection head.

To optimize the CL component, we employ the normalized temperature-scaled cross-entropy (NT-Xent) loss, which maximizes similarity between positive pairs while minimizing

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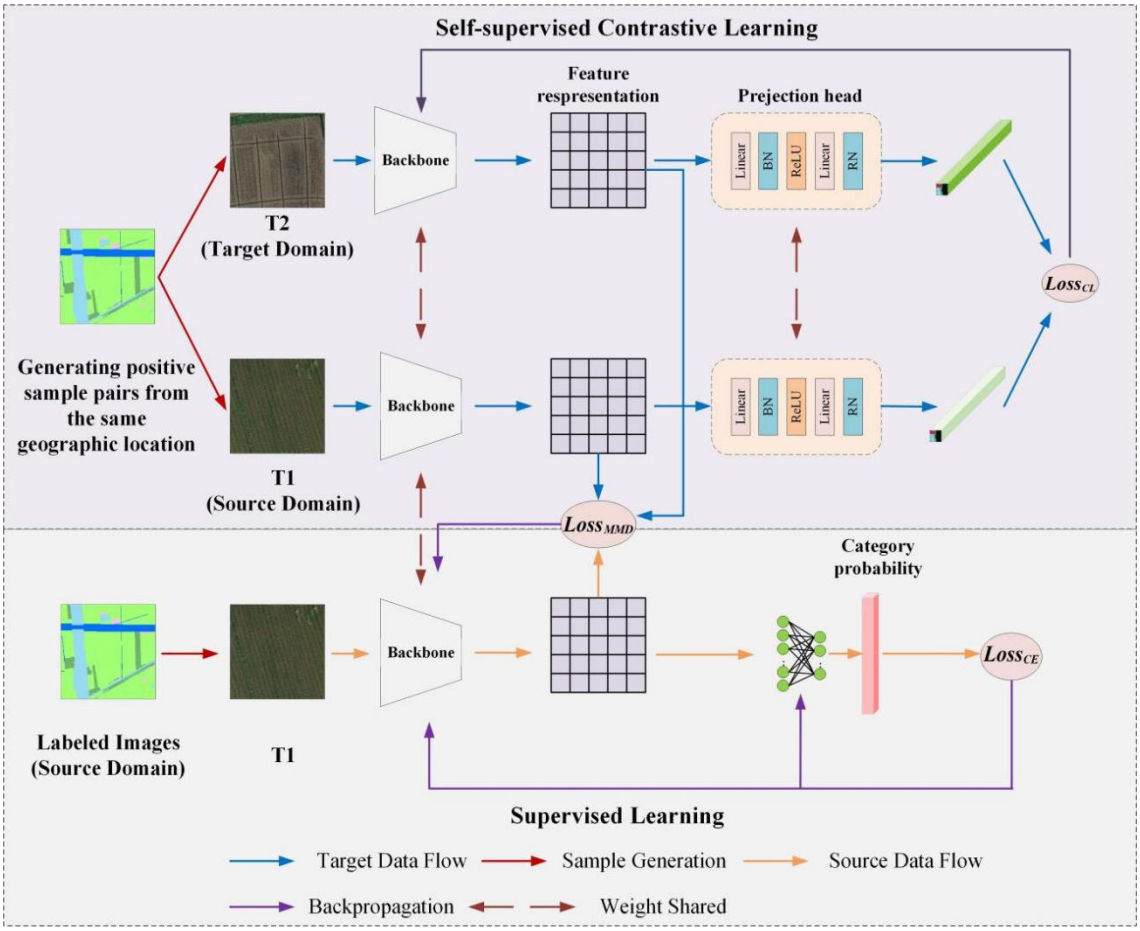


Fig. 4. Designed HCLF.

similarity with negative samples. For a positive pair (i, j), the contrastive loss is expressed as

$$\mathcal{L}_{CL} = -\log \frac{\exp\left(sim\left(z_i, z_j\right)/\tau\right)}{\sum_{k=1}^{2Y} 1_{[k\neq 1]} \exp\left(sim\left(z_i, z_k\right)/\tau\right)} \tag{9}$$

where $sim(\cdot)$ denotes the cosine similarity function, Y is the batch size, τ is the temperature parameter, and $1_{[k\neq 1]}$ is an indicator function equaling 1 if $k \neq i$ and 0 otherwise.

To further mitigate temporal domain discrepancies, an MMD loss [57] is introduced to explicitly minimize the distributional distance between features from different temporal domains in the latent space. By jointly optimizing the CL objective and the MMD-based alignment loss, the HCLF enforces temporal feature alignment, thereby improving representation consistency and enhancing generalization in cross-temporal semantic classification.

1) Fourier Pyramid Convolution Network (FPCNet): Fourier-domain transformations offer distinct advantages for CD in multimodal and cross-temporal remote sensing imagery, particularly in enhancing robustness to appearance variations. To overcome the limitations of conventional CNN backbones in cross-temporal feature representation, we replace the standard encoder with an FPCNet, as illustrated in Fig. 5. FPCNet takes RGB images as input. Its encoder begins with a stem block consisting of stacked 3×3 convolutional layers with a stride 1, each followed by batch normalization (BN) and Gaussian error linear unit (GELU) activation. This stem maps the input image into a high-dimensional feature space, suppressing color variations while preserving discriminative structural information relevant to SCD. A multiscale hybrid encoder architecture is then adopted to jointly optimize local detail extraction and global context modeling.

For high-resolution feature maps, a Fourier neural encoder (FNE) block (see Fig. 5) is employed to extract spatial–spectral representations, leveraging the global modeling capability of Fourier-domain operators for long-range dependency capture. For low-resolution feature maps, a lightweight ResNet structure with small convolutional kernels is used to extract fine-grained and weak semantic features. This hybrid design enables FPCNet to balance computational efficiency with strong representation capacity across scales.

Fig. 6 visualizes the spatial–spectral feature representations at different encoder dimensions. From left to right, the first six feature maps after the FNE block are shown. From top to bottom, the maps correspond to spatial-domain features, amplitude maps, and phase maps in the spectral domain. These visualizations reveal the synchronous generation of spatial and spectral features in FPCNet. The amplitude maps indicate that image energy is concentrated at specific frequencies, while frequency-domain changes induced by spatial

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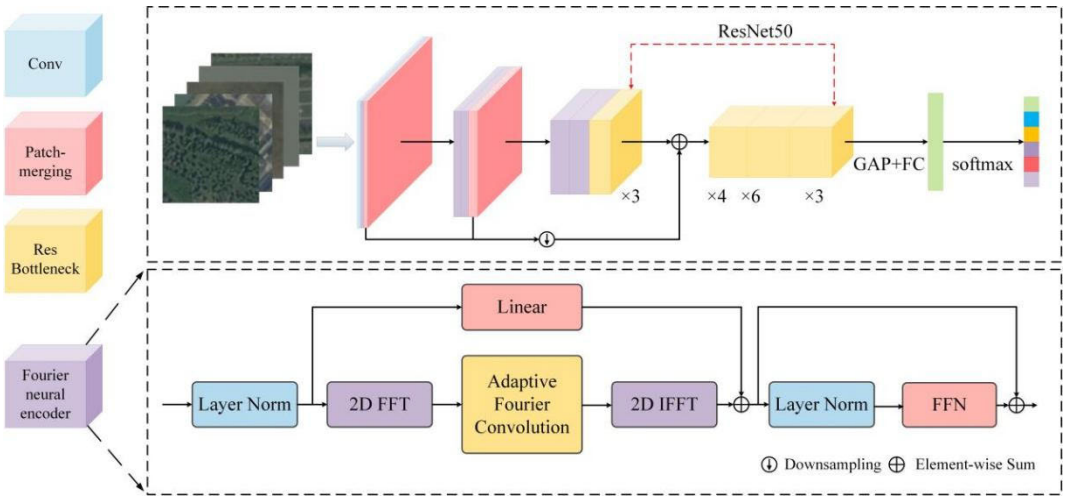


Fig. 5. Structure of the FPCNet used as the backbone feature extractor in HCLF.

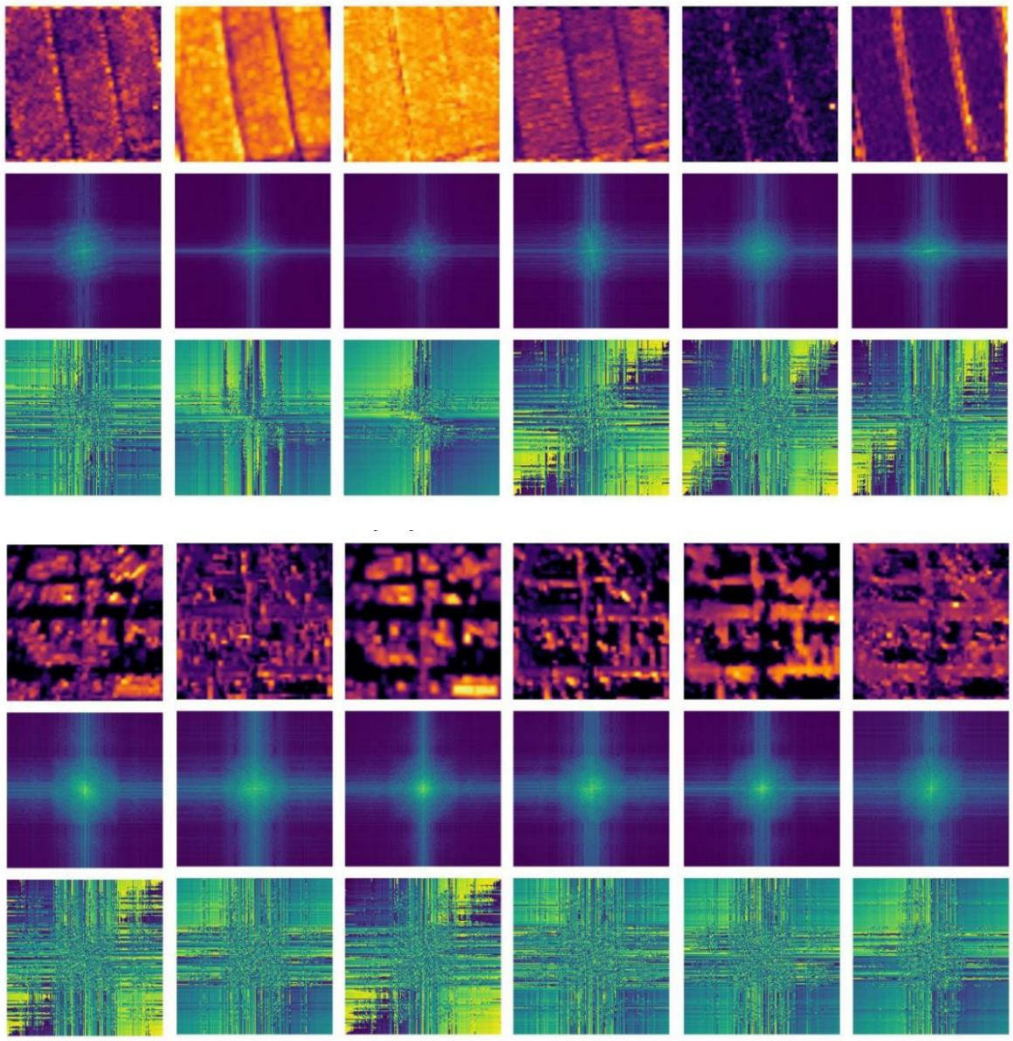


Fig. 6. Feature maps illustrating the spatial–spectral representations across different encoder dimensions after the FNE block.

differences remain stable. This confirms that Fourier spectrum transformations enhance the extraction of invariant features across images.

2) FNE and Fourier Asymmetric Convolution Layer: Leveraging the transformer encoder architecture and the demonstrated effectiveness of Fourier neural encoding in

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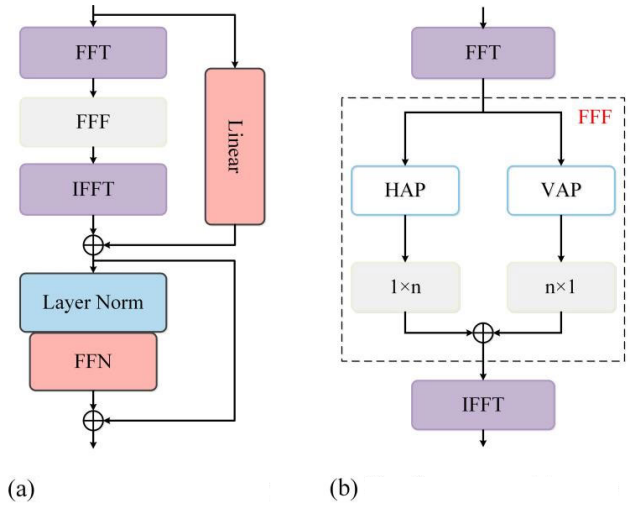


Fig. 7. (a) Structure of the FNE. (b) Structure of the Fourier asymmetric convolution layer.

vision tasks, and inspired by [13], we design a novel FNE module operating in the spatial–spectral domain, as illustrated in Fig. 7(a). The FNE comprises three main components: a Fourier neural operator using flexible Fourier filters (FFFs) in the spectral domain, a feed-forward network (FFN), and a linear transformation layer in the spatial domain. Here, FFT and IFFT denote the 2-D fast Fourier transform and its inverse, respectively. Specifically, the FFT projects spatial-domain feature maps into the frequency domain, enabling explicit modeling of global spectral characteristics. Subsequently, the IFFT transforms the enhanced frequency-domain features back into the spatial domain, ensuring compatibility with subsequent processing stages and facilitating visual interpretation.

Although FFF has demonstrated strong representational capacity in visual tasks, its large number of trainable parameters leads to increased computational cost and memory consumption. In contrast, asymmetric convolutions have been shown to preserve accuracy while reducing model complexity and improving efficiency [13]. Building on this observation, we propose a flexible separated scheme in the Fourier domain. This scheme projects spectral features along orthogonal directions and applies asymmetric convolutions independently along each direction, thereby achieving an effective tradeoff between parameter efficiency and feature extraction capability. As shown in Fig. 7(b), we integrate asymmetric convolution with Fourier convolution to construct a flexible Fourier asymmetric convolution (FAC) layer. The FAC layer comprises three lightweight components: vertical and horizontal average pooling for spectral projection, asymmetric convolution for extracting fine-grained spectral features in the frequency domain, and a feature aggregation operation.

a) Fourier asymmetric convolution: Given an input feature tensor $\boldsymbol{X} \in \mathbb{C}^{C\times H\times W}$, two directional representations are first derived via average pooling along orthogonal spatial directions to decouple dominant frequency responses. Specifically, horizontal and vertical average pooling are applied as

$$\begin{aligned} \boldsymbol{X}_H &= \mathrm{HAP}(\boldsymbol{X}) \\ \boldsymbol{X}_V &= \mathrm{VAP}(\boldsymbol{X}) \end{aligned} \tag{10}$$

where HAP and VAP denote horizontal and vertical average pooling, respectively. As a result, $\boldsymbol{X}_H \in \mathbb{C}^{C\times 1\times W}$ captures frequency variations along the horizontal axis, while $\boldsymbol{X}_V \in \mathbb{C}^{C\times H\times 1}$ captures variations along the vertical axis. This directional projection replaces the full spectral tensor and effectively reduces the frequency dimensionality while preserving dominant directional information. Subsequently, the compressed feature maps are processed using independent asymmetric convolution kernels aligned with their respective spatial directions, enabling efficient extraction of direction-specific spectral features. As illustrated in Fig. 7(b), the asymmetric convolution in the Fourier domain is formulated as

$$\boldsymbol{X}_F^H = Conv\mathrm{H}(\boldsymbol{X}_H) \tag{11}$$

$$\boldsymbol{X}_F^V = Conv\mathrm{V}(\boldsymbol{X}_V) \tag{12}$$

where $\boldsymbol{X}_F^H \in \mathbb{C}^{C\times 1\times W}$ and $\boldsymbol{X}_F^V \in \mathbb{C}^{C\times H\times 1}$ denote the extracted complex-valued features along the horizontal and vertical axes, respectively. $Conv\mathrm{H}(\cdot)$ and $Conv\mathrm{V}(\cdot)$ are asymmetric complex-valued convolution operators with kernel sizes $1\times n$ and $n\times 1$. The parameter n is selected to align with the convolutional kernel configurations in ResNet-50 and is therefore stage-dependent. Specifically, a value of n equal to 7 is adopted in the initial convolution stage, whereas a value of 3 is used in the residual stages.

b) Feature aggregation: To reconstruct a full spatial–spectral representation, the directional features are fused through element-wise addition with implicit broadcasting

$$\boldsymbol{X}_F = \boldsymbol{X}_F^H + \boldsymbol{X}_F^V \tag{13}$$

where $\boldsymbol{X}_F \in \mathbb{C}^{C\times H\times W}$ is the aggregated output feature tensor. The addition operator implicitly applies broadcasting (as implemented in PyTorch) to align tensor dimensions, enabling efficient fusion of complementary directional information while maintaining a compact parameterization.

3) Adaptive Fourier Convolution (AFC) Layer: Building upon the FAC layer, we propose an AFC layer, which incorporates a channel-wise meta-attention (CMA) module and an adaptive multiscale fusion (AMF) module, as illustrated in Fig. 8. This design enables the network to enhance semantic features adaptively while aligning and fusing multiscale information consistently. The CMA layer leverages attention mechanisms to recalibrate channel information, while the fusion module learns optimal combination parameters for multiscale feature maps. To improve robustness and generalization, we employ a hybrid regularization strategy: BN and Hardwish activation in the FAC layer, and layer normalization (LayerNorm) with Sigmoid activation in the MA layer.

Inspired by the squeeze-and-excitation mechanism [58], the CMA module adaptively models global contextual information to enhance semantically meaningful channels while suppressing less informative ones. To address challenges in remote sensing imagery, such as large-scale variations, complex backgrounds, and diverse appearances, we adopt spatial pyramid max pooling (SPMP) and spatial pyramid average pooling (SPAP). SPAP captures fundamental spectral characteristics while attenuating extreme responses, whereas SPMP

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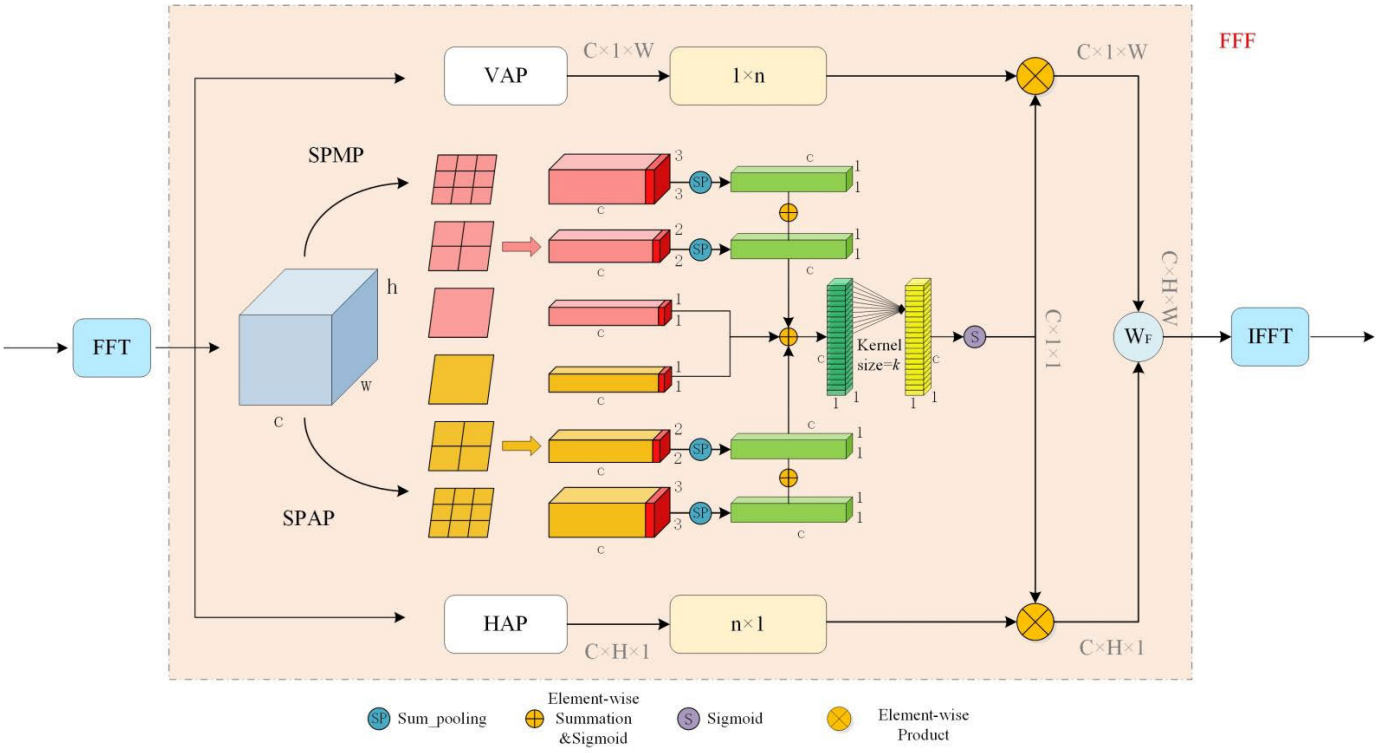


Fig. 8. Structure of the AFC layer.

emphasizes salient spectral features. The pooled multiscale features are then fused to recalibrate channel-wise importance. In implementation, LayerNorm is applied along the channel dimension for normalization, followed by a 1×1 convolution to promote model generalization. The Sigmoid activation function $\sigma(\cdot)$ is then used to form nonlinear features, thereby suppressing nonsemantic features while simultaneously enhancing semantic channels.

a) Meta-attention layer: For CNNs, channels in the feature map represent different features, and focusing on key channel features benefits model performance. Therefore, the two aforementioned types of global spatial features are fused to recalibrate the channel features, enabling the separated convolution layers to enhance semantic channel components and learn efficient multiscale feature representations, as shown in Fig. 8. First, for the input feature $\boldsymbol{X} \in \mathbb{C}^{C\times H\times W}$, two types of spatial pyramid pooling (SPP) operations are performed in parallel to obtain multiscale pooled features. The SPP multiscale pooled features for scales 1×1, 2×2, and 3×3 can be denoted as $\boldsymbol{X}^I_{max}$ and $\boldsymbol{X}^I_{avg}$, where $I \in \{1, 2, 3\}$. Subsequently, $\boldsymbol{X}^I_{max}$ and $\boldsymbol{X}^I_{avg}$ are down-sampled via sum pooling to generate $\hat{\boldsymbol{X}}^I_{max}$ and $\hat{\boldsymbol{X}}^I_{avg}$ of size $1 \times 1 \times C$, reducing computational overhead. Then, $\hat{\boldsymbol{X}}^I_{max}$ and $\hat{\boldsymbol{X}}^I_{avg}$ are aggregated via element-wise summation. The aggregation weight $\hat{W}$ is calculated as follows:

$$\hat{W}_c = \sigma\left(\sum_I \sigma\left(\hat{\boldsymbol{X}}^I_{max^I c}\right) + \sum_I \sigma\left(\hat{\boldsymbol{X}}^I_{avg^I c}\right)\right) \tag{14}$$

where $\hat{\boldsymbol{X}}^I_{max^I c}$ and $\hat{\boldsymbol{X}}^I_{avg^I c}$ are the down-sampled feature tensors for the cth channel, $\hat{W}_c$ is the cth channel of $\hat{W}$, and $\sigma(\cdot)$ denotes the sigmoid function.

A squeeze-and-excitation-like CMA layer is then defined as

$$\boldsymbol{X}^C_F = \sigma\left(LN\left(Conv_k\left(\hat{W}\right)\right)\right) \tag{15}$$

$$k = \left|\frac{log_2(C)}{\gamma} + \frac{b}{\gamma}\right|_{odd} \tag{16}$$

where $LN(\cdot)$ denotes complex LayerNorm, $Conv_k(\cdot)$ denotes a 1-D convolution with kernel size k, $|t|_{odd}$ denotes rounding to the nearest odd integer, and $\sigma(\cdot)$ again denotes the sigmoid function. In our experiments, γ and b are set to 2 and 1, respectively.

b) Adaptive multiscale fusion: Traditional feature aggregation methods, such as tensor addition or concatenation, have inherent limitations: 1) features from different scales may not be perfectly aligned and 2) features with differing dimensions cannot be semantically aggregated consistently. To address these issues, we propose a parameter learning-based AMF module [13]. The module recalibrates multiscale feature maps by learning adaptive fusion parameters, enabling better alignment of features across scales and more consistent semantic aggregation. Specifically, multiscale features processed by the FAC are first refined through the CMA layer. The FPCNet then adaptively calibrates the fusion weights of these features and optimizes the combined feature representation using channel attention. This mechanism enhances the network's resilience to feature noise and outliers, effectively accommodating perturbations introduced by cross-temporal shifts. The AMF layer, illustrated in Fig. 8, is formulated as

$$\mathbf{X}^{'}_F = \mathrm{W}^1_F \odot \left(\mathbf{X}^C_F \odot \mathbf{X}^V_F\right) + \mathrm{W}^2_F \odot \left(\mathbf{X}^C_F \odot \mathbf{X}^H_F\right) \tag{17}$$

where $\odot$ denotes element-wise multiplication, $\boldsymbol{X}^{'}_F \in \mathbb{C}^{C\times H\times W}$ is the output tensor, and $W^1_F \in \mathbb{C}^{C\times1\times W}$ and $W^2_F \in \mathbb{C}^{C\times H\times1}$ are

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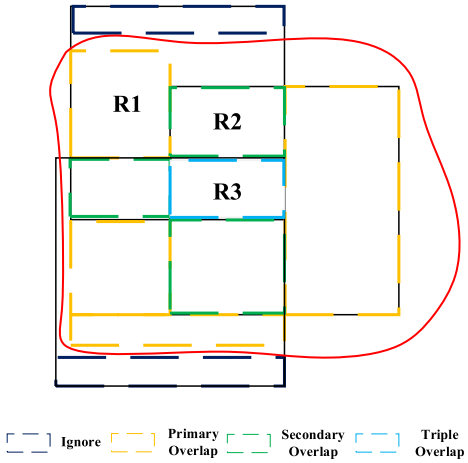


Fig. 9. Schematic of vector parcel attribute discrimination.

learnable parameters for adaptively calibrating the weights of the vertical and horizontal features, respectively.

C. Postclassification SCD

We propose a semantic-level postclassification CD solution, as illustrated in Fig. 9. Let $\{P_N^i\}$ be the set of image patches (i.e., scene samples) generated from the segmented region R_i, which serves as the postclassification processing unit, and N be the number of image patches. For example, when $i = 3$, $\{P_N^3\}$ represents the set of image patches formed by the red bounding boxes in Fig. 9. Let y_N^i denote the class attribute confidence for P_N^i. y_N^i can be calculated using the formula $y_N^i = countifs\{x\}$, where $\{x\}$ represents the set of class attribute values. The class attribute of P_N^i, denoted as $X_{P_N^i}$, is then assigned based on the maximum confidence rule: $X_{P_N^i} = argmax\{y_N^i\}$.

Let Y be the class attribute value of the vector parcel. Y can be calculated using the formula $Y = argmax\{S_x\}$, where S_x is the area corresponding to the class attribute value x, and x here represents the class attribute of different processing units after the postclassification processing. The CD decision for the vector parcel is performed using a threshold k, defined as

$$k = \frac{S_Y}{S_{total}} \times 100\% \tag{18}$$

where S_Y is the area corresponding to the assigned class Y, and S_{total} is the total area of the vector parcel. In our experiments, if $k \geq 90\%$, the parcel is assigned the class attribute Y. SCD is ultimately achieved by overlaying the classified parcels with the original vector parcels and comparing their attributes.

IV. Experiments

A. Experimental Data

The study areas include two real-world sites in Jiangsu Province, China, with corresponding imagery and vector parcel data derived from the Third National Land Survey. The vector parcel data were manually delineated during the initial (pre-event) time period.

1) Wuxi Dataset: This dataset comprises UAV-acquired RGB imagery over two periods, along with land-cover vector parcel data corresponding to the first-period imagery. The spatial resolution is 0.2 m, covering an area of 324.88 km^2, as illustrated in Fig. 10. Specifically, Fig. 10(a) and (b) displays the vector parcel data and the corresponding UAV imagery from February 2022, while Fig. 10(c) shows UAV imagery from July 2023 for the same region. The land-cover types primarily include buildings, forests, water bodies, cropland, and roads. Considering the high image resolution, the standard sample size was set to 224 pixels, with a minimum size of 64 pixels. Consequently, the final multiscale samples range from 224 × 224 to 64 × 64 pixels.

2) Nantong Dataset: Imagery for this area was acquired from the Beijing-2 satellite, with a spatial resolution of 0.8 m, covering 184 km^2, as shown in Fig. 11. Fig. 11(a) and (b) presents the vector parcel data and corresponding high-resolution imagery from October 2022, while Fig. 11(c) shows imagery from November 2023. Land-cover types include buildings, industrial land, forest, water bodies, cropland, and roads. The standard sample size was set to 156 pixels, with a minimum size of 48 pixels, resulting in multiscale samples ranging from 156 × 156 to 48 × 48 pixels.

3) Experimental Setup: All experiments were conducted on a workstation equipped with 64-GB RAM and an Intel Xeon CPU E-2186M @ 2.9 GHz. Model training was performed using an NVIDIA GeForce RTX 3090 Ti GPU with 24 GB of memory. The stochastic gradient descent (SGD) optimizer was employed with a momentum of 0.9 and a weight decay of 10^{-5}. The initial learning rate was set to 0.001 and decayed following a cosine annealing schedule. The batch size was fixed at 64.

B. Image Segmentation and Sample Generation

Considering the performance limitations of the experimental hardware, the original images and vector parcels of the selected study areas were first cropped into subsets not exceeding 3100 × 2500 pixels, with a cropping overlap rate of 15%. This strategy ensured comprehensive coverage and effective detection of vector parcels at the boundaries of each subset. Using the vector parcel boundaries from the pre-event phase, the proposed BPSLIC superpixel segmentation was applied to both temporal images, producing the initial training set and prediction set. To evaluate the effectiveness of BPSLIC, it was compared with the traditional SLIC method. Figs. 12 and 13 show the segmentation results of BPSLIC on partial regions of the Wuxi and Nantong datasets, respectively. The results demonstrate that, constrained by prior vector boundaries, BPSLIC generates more uniform and compact superpixels, with superior alignment to object boundaries such as rivers, buildings, and roads. This improved boundary fitting is advantageous for subsequent sample generation.

To quantitatively assess segmentation performance, a specific region within the Wuxi dataset was selected, and boundary recall [59] was adopted as the evaluation metric. A higher boundary recall indicates closer agreement between the segmentation boundaries and actual vector parcel boundaries. Fig. 14 presents a quantitative comparison between BPSLIC and SLIC, showing that BPSLIC consistently achieves higher boundary recall values. Given that SLIC is already a high-performing algorithm, this comparison confirms that BPSLIC

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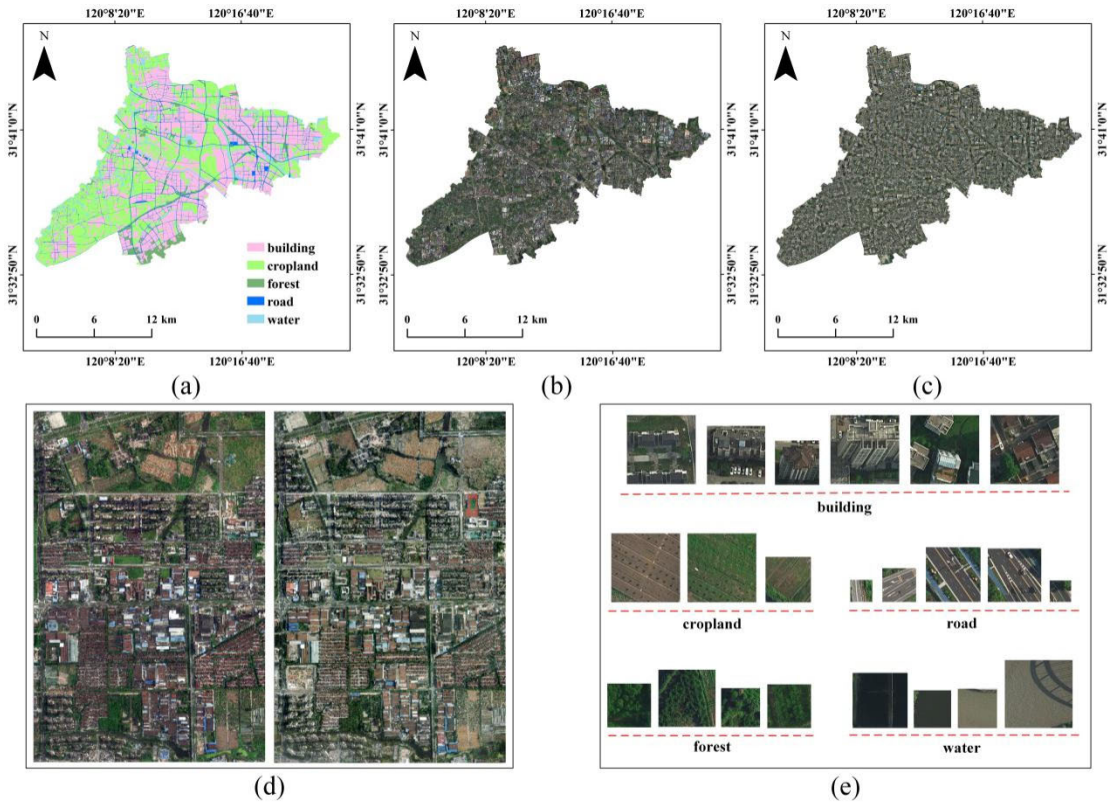


Fig. 10. Wuxi dataset collected from Huishan District, Wuxi City, Jiangsu Province, China. (a) Vector parcel from February 2022 corresponding to the contemporaneous UAV imagery. (b) UAV imagery from February 2022. (c) UAV imagery from July 2023. (d) Detailed view of the bi-temporal imagery. (e) Examples of land-cover samples from the dataset.

effectively improves boundary adherence, achieving near-optimal results rapidly and without heuristic guidance, thus providing instructive value for other studies.

After multiscale segmentation, the adaptive cropping strategy described in Section III-A was employed to generate the initial sample sets for both study areas. For categories with insufficient samples, data augmentation techniques such as rotation, scaling, translation, and cropping were applied. Finally, the proposed denoising framework was used to obtain the final training-sample set. Examples of land-cover samples from the Wuxi and Nantong datasets are shown in Figs. 10(e) and 11(e), respectively, and the label distributions of the constructed sample sets are presented in Fig. 15.

C. Scene Classification

The proposed cross-temporal SCD method operates as a postclassification CD approach, relying on image classification results. It uses segmented and cropped images from two temporal phases: the T1 phase for training and the T2 phase for prediction. Classification categories are defined by the land-use attributes of vector parcels. This strategy not only ensures that samples are relevant to the study areas but also captures detailed "from-to" change information at the sample level, enabling refined characterization of land-use and land-cover changes.

Building on this framework, we propose an HCLF specifically tailored for cross-temporal classification. To validate its performance, we compared HCLF against mainstream CL models and traditional supervised learning models, using image samples from different temporal phases as training and prediction sets. Various backbone network architectures were also explored to identify the optimal configuration. Overall accuracy (OA) and Kappa coefficient were adopted as evaluation metrics, with results presented in Table I.

Compared with supervised classification models such as ResNet-50, WaveMLP-S [60], ConvNeXt [12], and PatchConvNet [61], HCLF demonstrates superior performance by effectively integrating the strengths of supervised learning and CL. For state-of-the-art CL models, including SwAV [62], SimSiam [63], and MMCR [64], all using ResNet-50 as the backbone, HCLF still outperforms them, achieving 92.05% OA and a Kappa of 0.902 on the Wuxi dataset, and 88.51% OA and a Kappa of 0.879 on the Nantong dataset. In addition, compared with the semi-supervised cross-domain CL framework SCCLF [65], HCLF achieves further accuracy improvements across both datasets, demonstrating its effectiveness in combining CL pretraining and fine-tuning with supervised learning.

The performance gains are more pronounced on the Wuxi dataset than on the Nantong dataset. This can be attributed to intrinsic dataset characteristics: Wuxi consists of high-resolution UAV imagery with a larger temporal gap between acquisitions (winter versus summer) and more complex landscape features, providing greater scope for HCLF to

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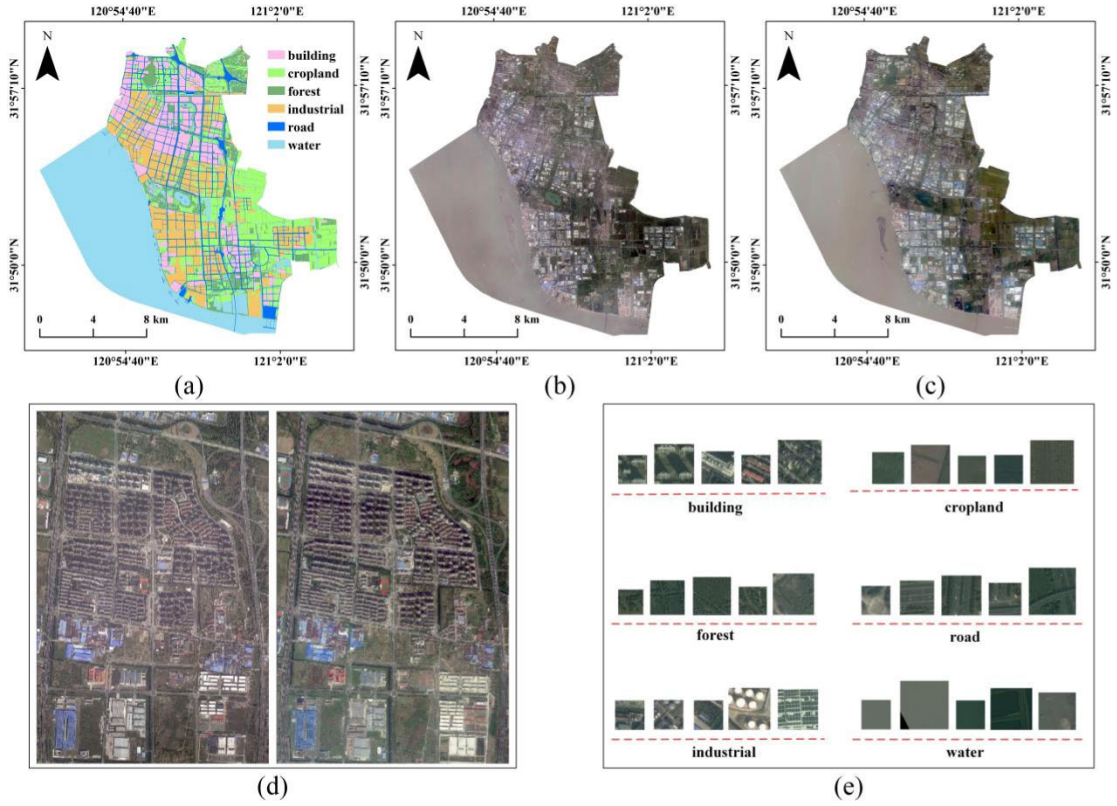


Fig. 11. Nantong dataset collected from the Economic and Technological Development Zone, Nantong City, Jiangsu Province, China. (a) Vector parcel from October 2022 corresponding to the contemporaneous high-resolution imagery. (b) High-resolution imagery from October 2022. (c) High-resolution imagery from November 2023. (d) Detailed view of the bi-temporal imagery. (e) Examples of land-cover samples from the dataset.

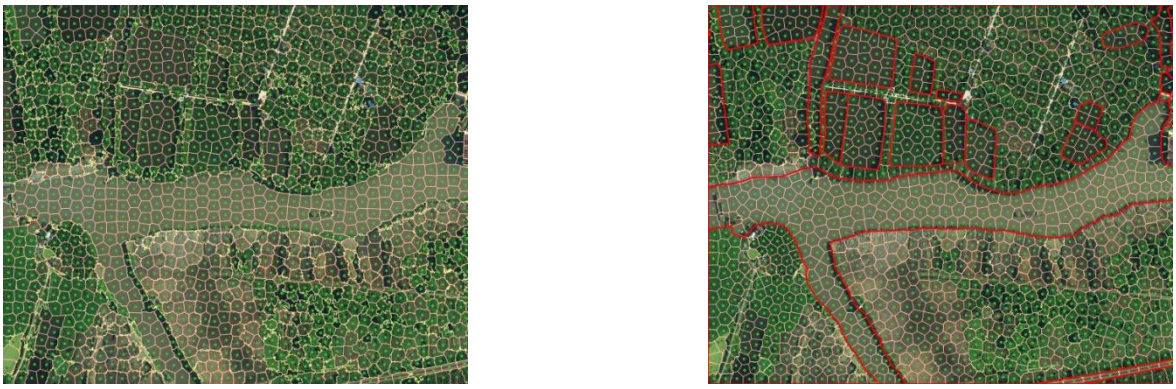


Fig. 12. Performance comparison of different superpixel segmentation algorithms in Huishan District, Wuxi, using the same number of seed points. (a) SLIC. (b) BPSLIC. Yellow boundaries indicate superpixel segments, while red boundaries represent vector parcel boundaries.

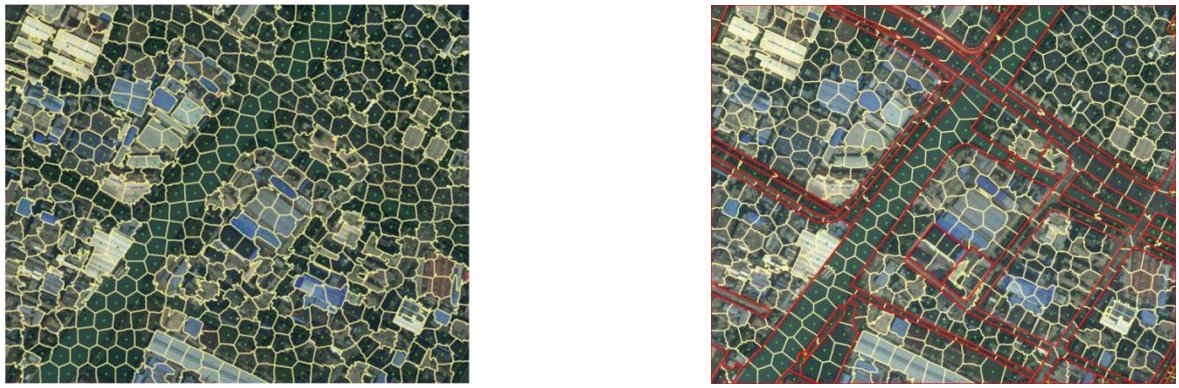
demonstrate its robustness. In contrast, the Nantong dataset consists of satellite imagery acquired in autumn for both phases, featuring more uniform or simpler characteristics, which reduces the margin for improvement. Dataset size, classification category differences, and annotation quality may also influence performance metrics.

Regarding computational efficiency, HCLF has 27.39-M parameters and 4.12-G FLOPs. Compared to the ResNet-50 baseline, this adds only ~2.78-M parameters while maintaining similar FLOPs, yet yields more than 8 percentage points improvement in OA across both study areas. This indicates that the proposed improvements enhance cross-domain learning performance without significant computational overhead, achieving a good balance between accuracy and efficiency.

D. CD Results

The proposed CD method employs vector parcels as the fundamental detection units and performs semantic-level change identification through a postclassification comparison strategy. The workflow proceeds as follows: first, the trained HCLF model performs fine-grained scene classification on samples within each vector parcel in the T2 imagery. Then, the

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(a) (b)

Fig. 13. Performance comparison of different superpixel segmentation algorithms in the Nantong Economic and Technological Development Zone using the same number of seed points. (a) SLIC. (b) BPSLIC. Yellow boundaries indicate superpixel segments, while red boundaries represent vector parcel boundaries.

TABLE I
CLASSIFICATION ACCURACY OF DIFFERENT MODELS IN THE STUDY AREAS

Model	Params(M)	FlOPs(G)	Wuxi		Nantong	
			OA(%)	kappa	OA(%)	kappa
ResNet-50	24.61	4.12	83.21	0.805	84.70	0.826
WaveMLP-S	30.73	4.53	86.65	0.841	85.85	0.841
ConvNeXt	27.81	4.52	85.38	0.829	86.58	0.852
PatchConvNet	26.44	4.06	87.92	0.854	85.25	0.846
SwAV	25.96	4.12	88.25	0.857	84.15	0.828
SimSiam	25.13	4.12	87.68	0.851	85.92	0.833
MMCR	25.79	4.12	89.47	0.889	86.28	0.849
SCCLF(ResNet-50)	25.63	4.12	90.55	0.890	87.76	0.864
SCCLF(EfficV2-S)	21.11	2.95	90.79	0.899	87.91	0.856
Ours(ResNet-50)	27.39	4.12	**92.05**	**0.902**	**88.51**	**0.879**

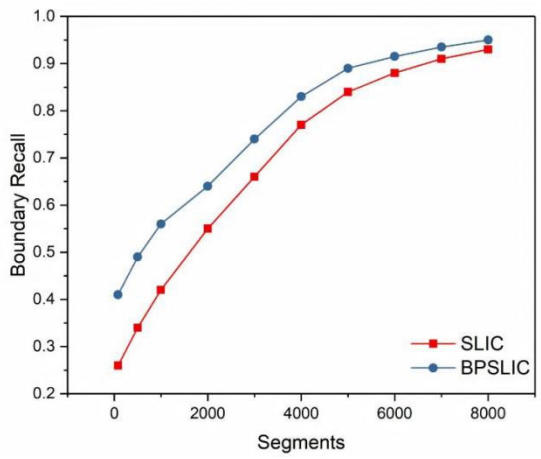


Fig. 14. Quantitative comparison of the BPSLIC and SLIC segmentation methods in a specific area of the Wuxi dataset.

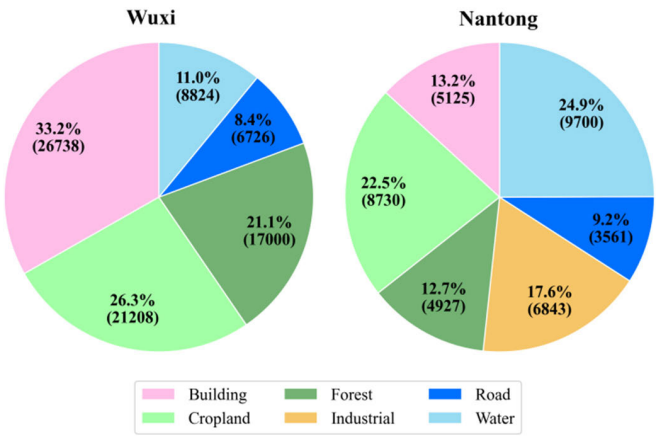


Fig. 15. Label distribution chart of the datasets.

classification results are combined with the original T1 vector attributes, and an overlay analysis is conducted using predefined rules to identify polygons undergoing semantic changes. To improve the rationality and consistency of semantic change identification, the following decision rules were established.

1) Sensitive Types: This category primarily includes artificial features such as buildings and industrial land. The determination threshold is set to 0. This means if any sample within a cropland polygon is predicted as a sensitive type, the entire polygon is deemed as changed.

2) Nonsensitive Types: This category mainly includes features such as roads, water bodies, and forests. The

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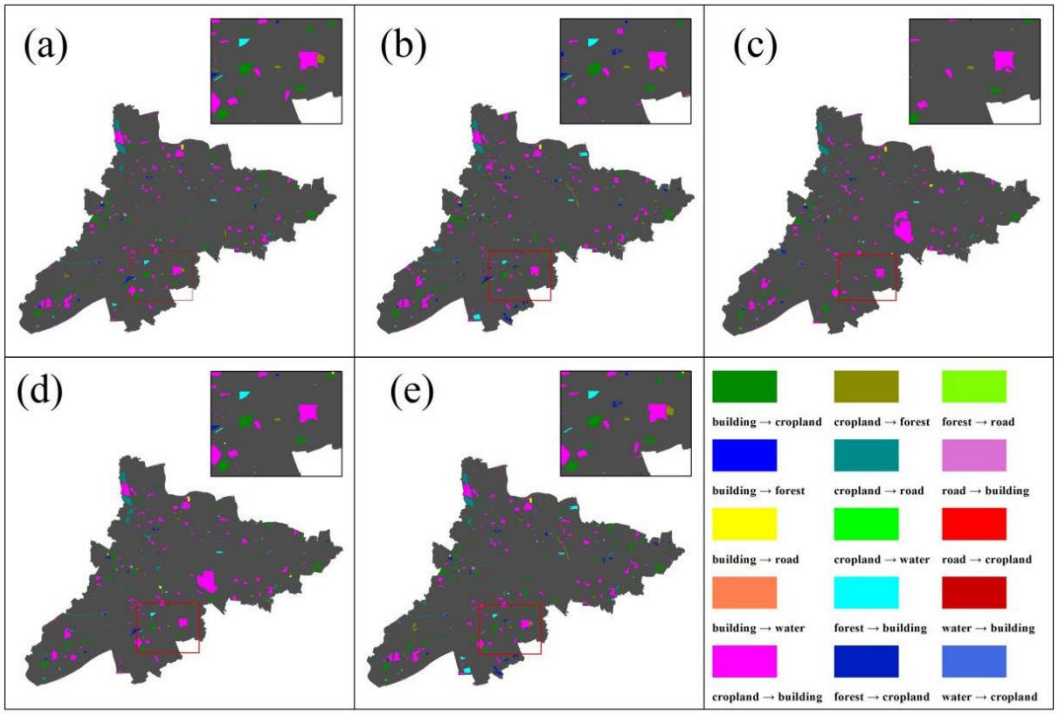


Fig. 16. Comparison of SCD visualization results for the Wuxi dataset. (a) Ground truth. (b) SLIC + without C^2-DPC + ResNet-50. (c) BPSLIC + without C^2-DPC + ResNet-50. (d) BPSLIC + C^2-DPC + ResNet-50. (e) BPSLIC + C^2-DPC + HCLF.

TABLE II
ACCURACY COMPARISON BEFORE AND AFTER SAMPLE PURIFICATION FOR THE WUXI DATASET

Training Set		Epoch					
		10	20	30	40	60	75
Initial	TA	0.8333	0.8612	0.8795	0.8759	0.8881	**0.8895**
Purified	TA	0.8412	0.8879	0.9076	0.9189	0.9235	**0.9205**

determination threshold is dynamically set between 0.1 and 0.3. This means if the proportion of samples predicted as nonsensitive types within a cropland polygon does not exceed the threshold, the polygon is considered unchanged.

To ensure the objectivity and reliability of accuracy assessment, the reference data were independently generated through expert visual interpretation prior to applying the proposed method. For postdetection accuracy evaluation, the detection results were compared against this pre-established reference dataset to count correct detections, false alarms, and missed detections. In accordance with the minimum mapping unit requirement of the national land change survey project, polygons smaller than 200 m^2 were removed. precision and recall were then calculated based on these statistics.

The experimental results indicate that the proposed method achieves high CD accuracy in both study areas. The Wuxi dataset obtained a precision of 88.27% and a recall of 77.71%, whereas the Nantong dataset achieved 77.21% and 75.42%, respectively, demonstrating the method's robustness and effectiveness for detecting nonagricultural land conversion in real-world settings. Figs. 16 and 17 provide visual comparisons of CD results for the Wuxi and Nantong datasets. In both figures, (a) presents the ground-truth reference changes, while (b)–(e) display detection results from different methodological combinations. The complete proposed pipeline (BPSLIC + C^2-DPC + HCLF) achieves the best overall performance, exhibiting superior boundary integrity, improved internal consistency, and more effective suppression of false changes. This further confirms the contribution and necessity of each module within the pipeline. Fig. 18 shows representative examples of SCD outcomes for both datasets. The land-use types in (a) and (c) exhibit transformations from cropland to construction land; (b) shows a transition from construction land to forest (reflecting demolition); and (d) shows cropland converted to industrial/mining land. Overall, the results demonstrate strong consistency with the reference data and satisfactory performance across diverse change scenarios.

V. DISCUSSION

A. Impact of Sample Denoising on Model Accuracy Improvement

To quantitatively assess the effectiveness of the proposed C^2-DPC denoising framework, we compared the training accuracy (TA) of models trained using the initial and purified sample sets across training epochs. As reported in Tables II (Wuxi) and III (Nantong), model performance improves

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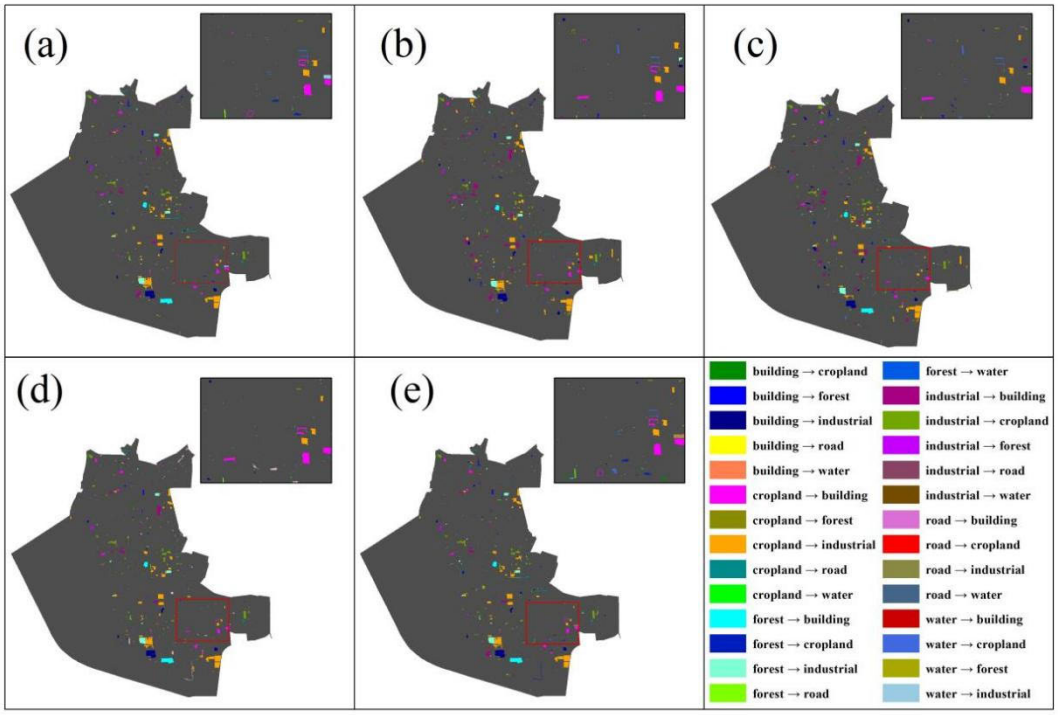


Fig. 17. Comparison of SCD visualization results for the Nantong dataset. (a) Ground truth. (b) SLIC + without C^2-DPC + ResNet-50. (c) BPSLIC + without C^2-DPC + ResNet-50. (d) BPSLIC + C^2-DPC + ResNet-50. (e) BPSLIC + C^2-DPC + HCLF.



Fig. 18. Examples of cross-temporal CD results. (a) and (b) Correspond to the Wuxi dataset, while (c) and (d) correspond to the Nantong dataset.

consistently and substantially after denoising. For the Wuxi dataset, final TA (epoch 75) improved from 0.8895 to 0.9205, an absolute gain of 3.10%. Improvements are even more pronounced for the Nantong dataset, where final TA (epoch 80) increased from 0.8420 to 0.8851, a gain of 4.31%.

An important observation is the accelerated convergence speed. Across both datasets, models trained with purified samples achieve higher accuracy at earlier epochs. For example, on the Nantong dataset, TA at epoch 20 increased from 0.8156 to 0.8733. This indicates that removing noisy or mislabeled samples reduces feature ambiguity, enabling the model to converge more rapidly to a superior solution. Accordingly, the denoising framework significantly enhances both final model accuracy and training efficiency, underscoring its role as a critical data preprocessing component.

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TABLE III
ACCURACY COMPARISON BEFORE AND AFTER SAMPLE PURIFICATION FOR THE NANTONG DATASET

Training Set		Epoch					
		10	20	30	40	60	80
Initial	TA	0.7634	0.8156	0.8324	0.8361	0.8401	**0.8420**
Purified	TA	0.8179	0.8733	0.8789	0.8815	0.8845	**0.8851**

TABLE IV
ABLATION STUDY RESULTS FOR DIFFERENT MODULES IN HCLF ON THE TWO DATASETS

Baseline	CL	$Loss_{MMD}$	AFC	Wuxi		Nantong	
				OA(%)	kappa	OA(%)	kappa
✓				83.47	0.817	84.71	0.828
✓			✓	84.28	0.838	85.58	0.835
✓	✓		✓	87.15	0.854	86.34	0.855
✓	✓	✓		90.93	0.891	87.69	0.864
✓	✓	✓	✓	**92.26**	**0.902**	**88.35**	**0.867**

TABLE V
ABLATION STUDY ON THE INTERNAL MODULES OF FNE

Baseline	FAC	CMA	AMF	Wuxi		Nantong	
				OA(%)	kappa	OA(%)	kappa
✓				90.93	0.891	87.69	0.864
✓	✓			91.88	0.899	88.17	0.866
✓	✓	✓		92.02	0.900	88.24	0.867
✓	✓	✓	✓	**92.26**	**0.902**	**88.35**	**0.867**

B. Ablation Study

A comprehensive ablation study was conducted to evaluate the contribution of each component of the proposed HCLF framework, with the results summarized in Table IV. Using ResNet-50 trained with C^2-DPC-purified samples as the baseline, overall accuracies of 83.47% and 84.71% were obtained on the Wuxi and Nantong datasets, respectively. Incorporating the AFC module resulted in moderate performance gains. Introducing CL led to more pronounced improvements, indicating the importance of explicitly modeling intersample relationships for robust feature representation. The most substantial gain was achieved by further integrating the MMD loss, which increased the OA to 90.93% on Wuxi and 87.69% on Nantong, highlighting the effectiveness of mitigating interregion domain discrepancies. The full configuration combining the baseline with CL, MMD loss, and AFC achieved the best performance, with OAs of 92.26% and 88.35% on the two datasets, respectively. These results confirm that each component contributes positively and that the complete framework enhances both discriminative capability and cross-domain generalization.

To further analyze the internal design of the AFC module within the FNE block, an additional ablation study was performed on its constituent components, including FAC, CMA, and AMF, as reported in Table V. Using a ResNet-50 model equipped with CL and MMD loss as the baseline, the introduction of FAC produced consistent performance improvements on both datasets, reflecting the benefits of spectral-domain feature enhancement. Specifically, FAC increased the OA by 0.95% on Wuxi and 0.48% on Nantong, demonstrating its effectiveness in cross-temporal CD. Further incorporating CMA yielded additional gains, while the complete AFC configuration integrating FAC, CMA, and AMF achieved the best overall performance across all evaluation metrics. These findings indicate that the proposed submodules are complementary and jointly contribute to improved cross-temporal SCD.

C. Visualization and Evaluation of Feature Alignment in the HCLF Framework

This section investigates whether time-invariant features are effectively extracted by the proposed HCLF framework and evaluates the degree of feature alignment across temporal domains. Kernel density estimation (KDE) [66] plots are employed to visualize and compare feature distributions before and after CL. Figs. 19 and 20 present the KDE-based feature visualizations for the Wuxi and Nantong datasets, respectively. The x-axis represents feature values, while the y-axis denotes the relative likelihood of observing a given feature value, as estimated by KDE. A higher density indicates a greater probability of occurrence.

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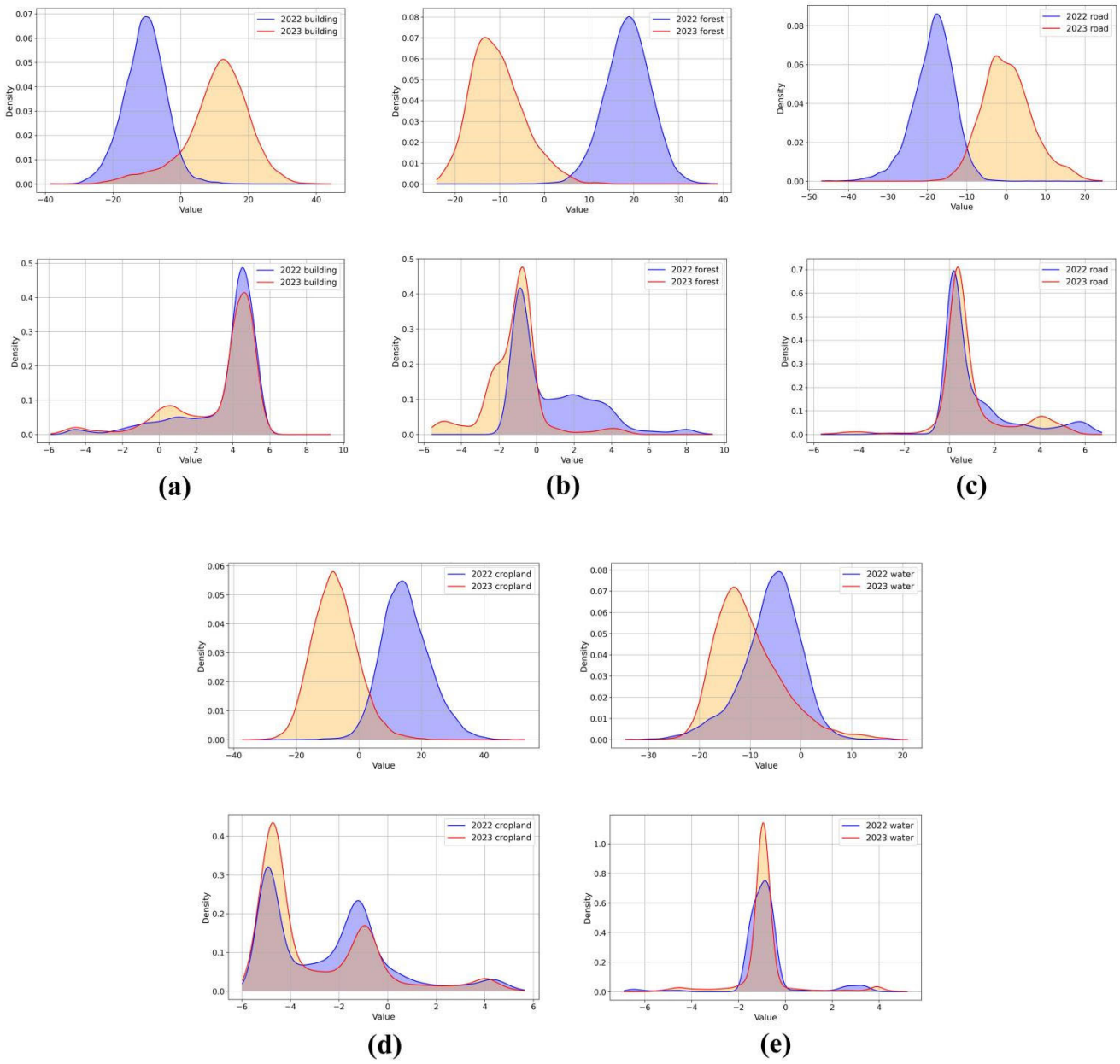


Fig. 19. KDE-based visualization of features extracted by different classifiers on the Wuxi dataset. The *X*-axis denotes feature values, and the *y*-axis represents the relative likelihood of observing a given feature value estimated by KDE. Upper subplots in (a)–(e) present the KDE distributions of features for building, forest, road, cropland, and water, respectively, without the feature alignment module. Lower subplots in (a)–(e) show the corresponding distributions after applying the HCLF feature alignment module.

In each subfigure, the upper part illustrates the feature distributions of different land-cover classes across temporal phases when directly using ResNet-50. The bimodal distributions with distinct colors indicate that land-cover features from different temporal phases exhibit divergent distributions. The lower part of each subfigure shows the features extracted by the HCLF framework after applying the domain feature alignment strategy. It is evident that, after incorporating the domain feature alignment module, features corresponding to the same land-cover classes from different temporal phases exhibit more similar and consistent distributions, with substantially increased overlap. These results demonstrate that the proposed HCLF framework is effective in extracting time-invariant deep features.

D. Limitations and Future Work

Although the proposed framework demonstrates encouraging performance, several limitations remain. First, the effectiveness of the vector-guided learning strategy depends on the quality and temporal consistency of auxiliary vector parcel data; inaccurate or outdated annotations may degrade denoising performance and consequently affect CD accuracy. Second, the vector-based parcel-level CD strategy, while facilitating result updating, is limited to identifying changed parcel extents and cannot provide fine-grained delineation. Third, the current framework is designed for bi-temporal image pairs, and extending it to multitemporal or long-term time-series scenarios may require further architectural modifications and more

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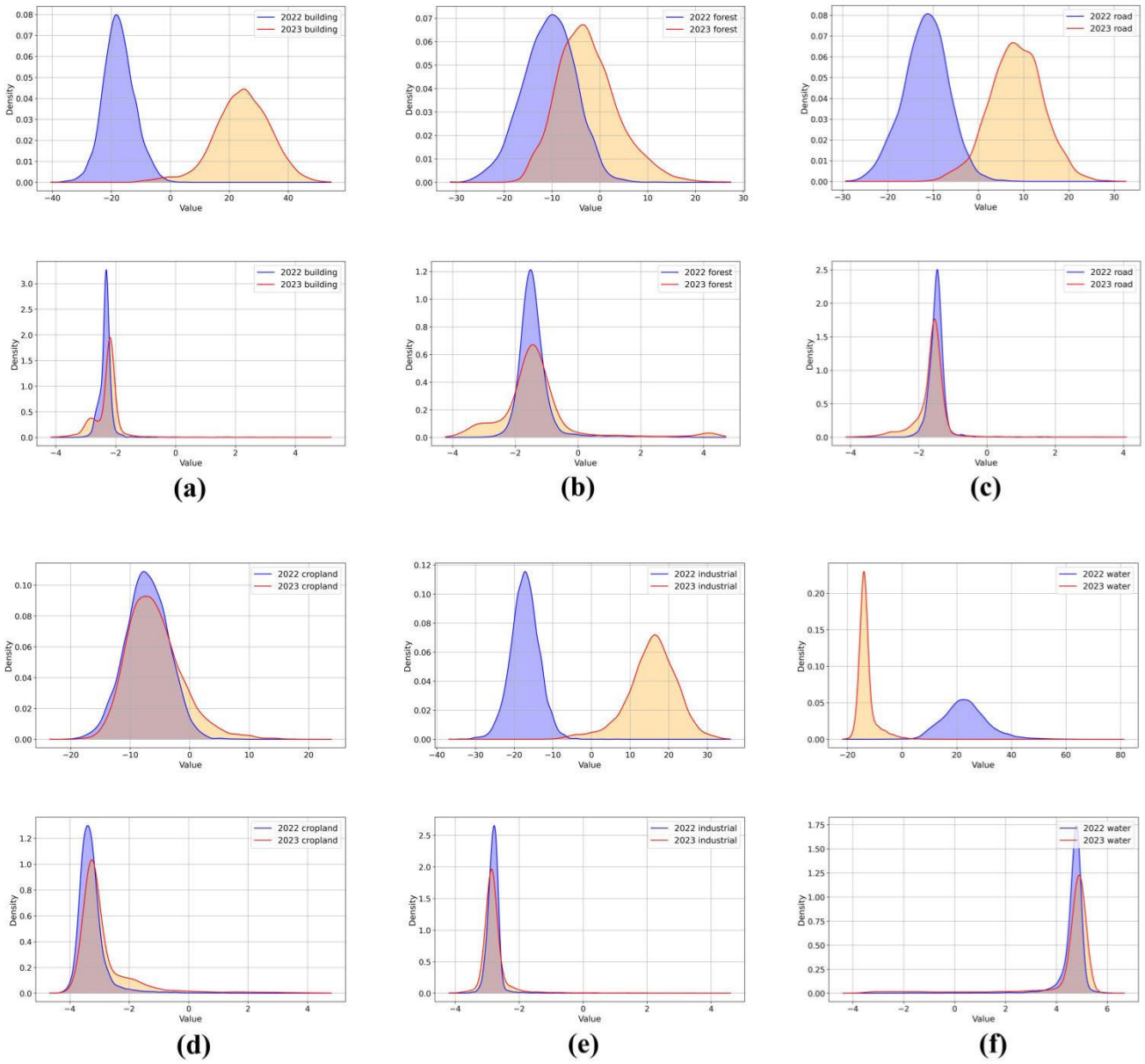


Fig. 20. KDE-based visualization of features extracted by different classifiers on the Nantong dataset. The *x*-axis denotes feature values, and the *y*-axis represents the relative likelihood of observing a given feature value estimated by KDE. Upper subplots in (a)–(f) present the KDE distributions of features for building, forest, road, cropland, industrial, and water, respectively, without the feature alignment module. Lower subplots in (a)–(f) show the corresponding distributions after applying the HCLF feature alignment module.

advanced temporal modeling. Finally, the rule-based change decision module improves interpretability and stability but relies on empirically determined thresholds, which may require adjustment when applied to regions with substantially different land-cover characteristics. Future work will investigate foundation segmentation models for finer delineation, lightweight frequency-domain representations, and broader validation on diverse multitemporal datasets.

VI. Conclusion

This study addresses the dual challenges of cross-temporal domain shift and the scarcity of high-quality labeled samples in remote sensing SCD by proposing a feature-level alignment framework that integrates vector parcel data with a CL strategy. By combining BPSLIC superpixel segmentation with adaptive cropping, the proposed approach enables automatic training-sample generation and effective denoising, substantially reducing dependence on labor-intensive manual annotation. The HCLF jointly exploits supervised learning, self-supervised CL, and an MMD-based feature alignment loss to explicitly align feature distributions across temporal phases, thereby improving robustness to variations in seasonality, illumination, and acquisition conditions. To further enhance invariant feature representation, an adaptive FPCNet (AFPCNet) is developed to facilitate interactions between spatial and spectral representations. Its multiscale fusion architecture captures both global contextual information and local

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structural details, enabling the learning of discriminative and temporally stable features in cross-temporal remote sensing imagery.

Experiments conducted in two real-world study areas, Huishan District in Wuxi and the Economic and Technological Development Zone in Nantong, demonstrate the effectiveness of the proposed method. In cross-temporal scene classification, the framework achieves OAs of 92.05% with a Kappa coefficient of 0.902 on the Wuxi dataset and 88.51% with a Kappa coefficient of 0.879 on the Nantong dataset, outperforming mainstream supervised and CL approaches. In the subsequent SCD task, the method attains precision and recall values of 88.27% and 77.71% in Wuxi, and 77.21% and 75.42% in Nantong, respectively, confirming its strong practical potential for automated and high-precision CD.

Future work will explore deeper integration of multimodal vector parcel data and investigate zero-shot or few-shot cross-domain learning strategies based on large language models (LLMs) or vision language models (VLMs) to further improve generalization and interpretability under complex scenarios with limited annotations.

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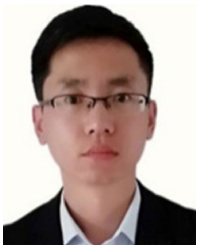
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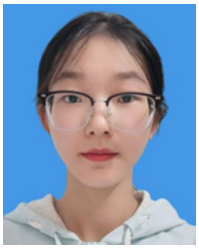
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